TILINGS AND MOSAICS

Regular, semiregular, aperiodic and Islamic tilings, coloured with colour theory

A tiling covers the plane with shapes, without gaps or overlaps. The same tiling changes character with its colouring: by kind of tile, by orientation, by mirror image, or so that neighbouring tiles always differ. Every page below draws the tilings of one Wikipedia article with several colourings; the colours come from the colour schemes of the RYB wheel or from traditional tile palettes. All of it is computed: lattices of polygons, duals, Penrose deflation, the hat metatiles, a search for Wang tilings and Hankin's method for star patterns.

Colour schemes

RYB schemes take hues at fixed steps of the 12-hue wheel from a base hue (here red); when a tiling needs more colours than a scheme has, tints and shades of the same hues are added. The named palettes copy traditional glazed tiles.


monochromatic (red)


analogous (red)


complementary (red)


split complementary (red)


triad (red)


tetrad (rectangle) (red)


square (red)


rainbow (red)


ramp, dark to light (red)


sweep of hues (red)


zellij palette


azulejo palette


girih palette


earth tones


greys

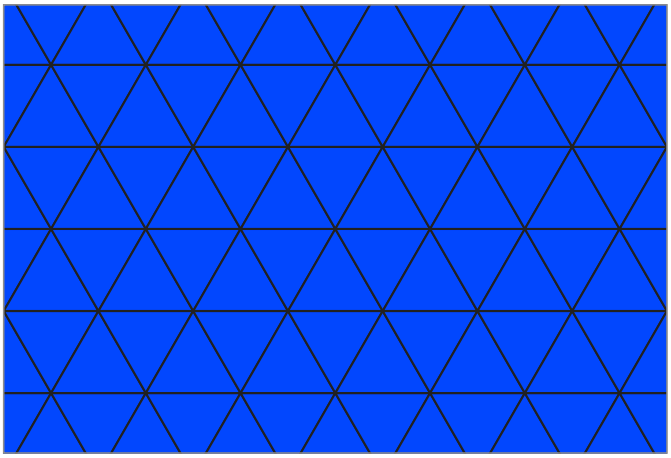
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Triangular tiling

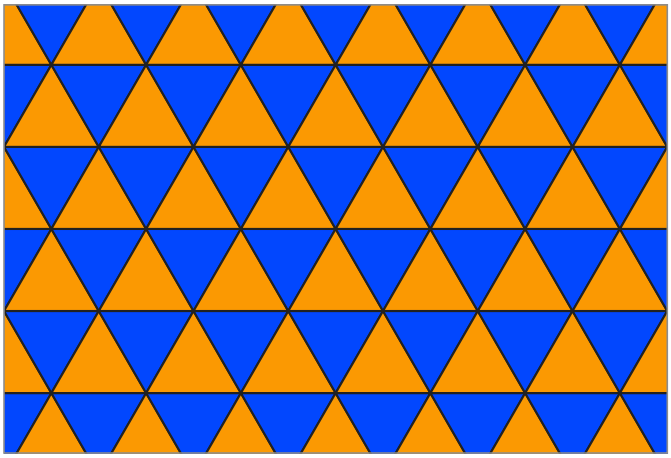
Wikipedia: Triangular tiling — en.wikipedia.org/wiki/Triangular_tiling

Six equilateral triangles meet at every vertex. Its uniform colourings are named by the colours around a vertex: 121212 alternates two colours, 121314 needs four. Cutting the triangles further gives the kisrhombille and kisdeltile tilings.



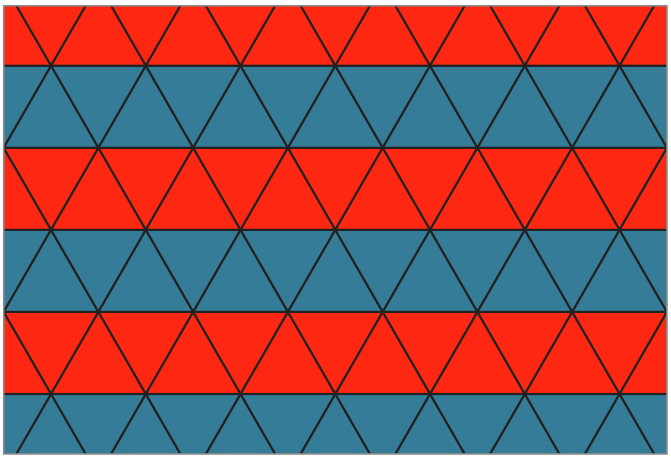
Regular triangular tiling {3,6}

Colouring: 111111 · monochromatic (blue)



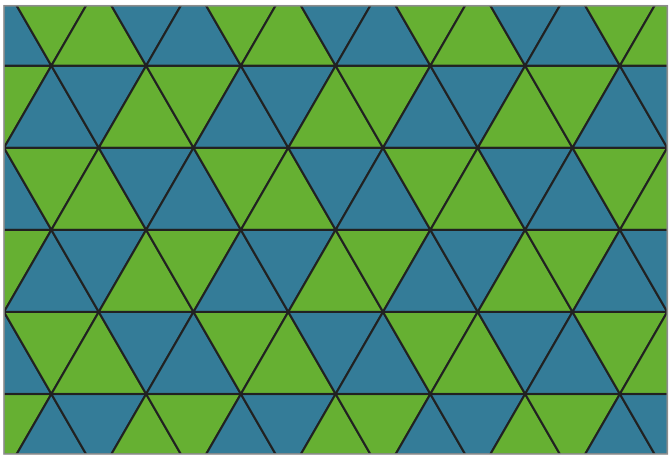
Uniform colouring 121212

Colouring: 121212 · complementary (orange)



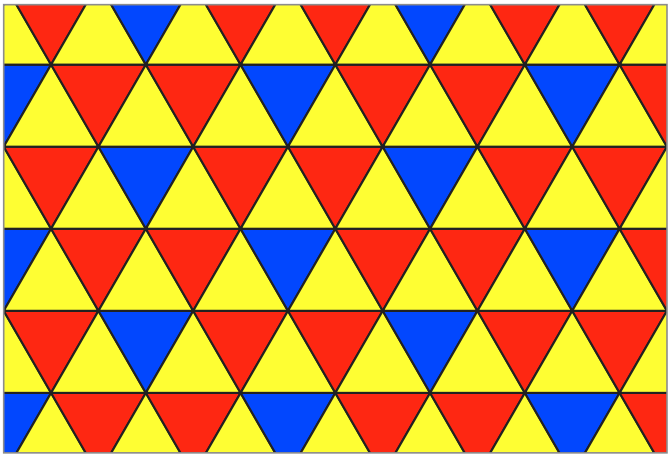
Uniform colouring 111222

Colouring: 111222 · split complementary (red)



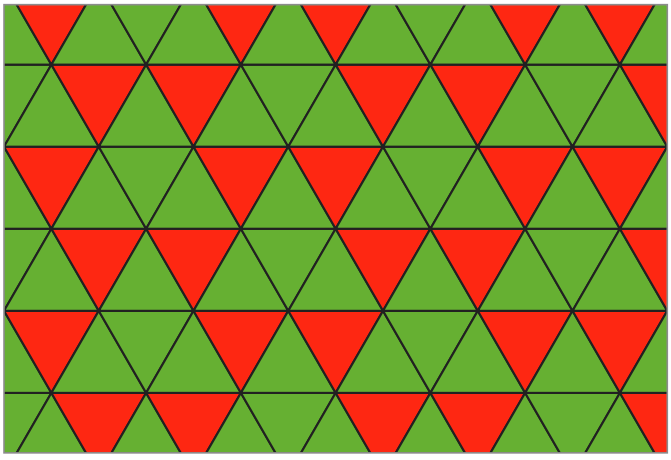
Uniform colouring 112122

Colouring: 112122 · analogous (blue-green)



Uniform colouring 121213

Colouring: 121213 · triad (yellow)



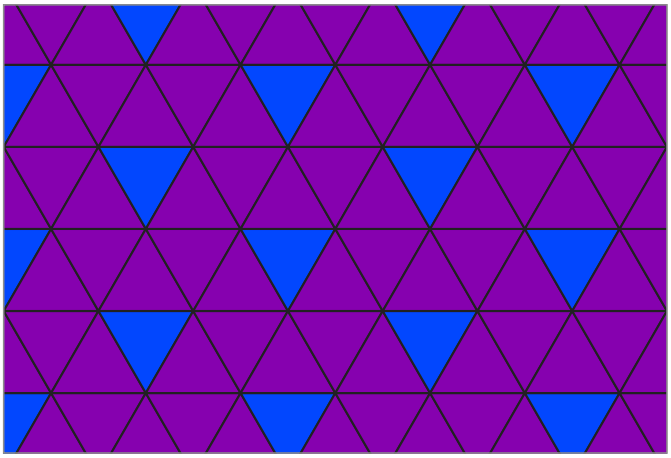
Uniform colouring 111212

Colouring: 111212 · complementary (green)

Triangular tiling (continued)

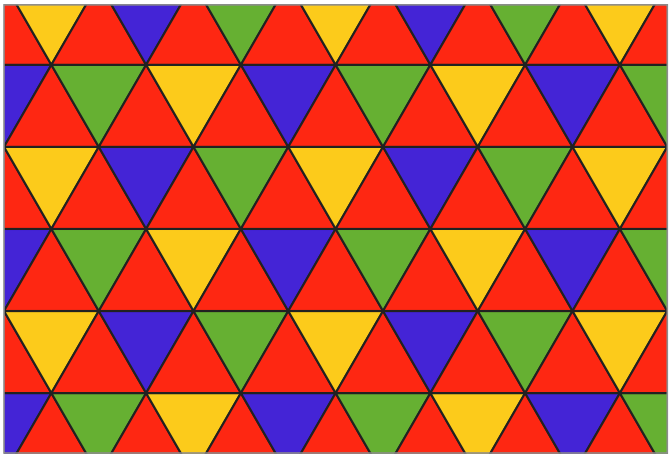
Wikipedia: Triangular tiling — en.wikipedia.org/wiki/Triangular_tiling

Six equilateral triangles meet at every vertex. Its uniform colourings are named by the colours around a vertex: 121212 alternates two colours, 121314 needs four. Cutting the triangles further gives the kisrhombille and kisdeltile tilings.



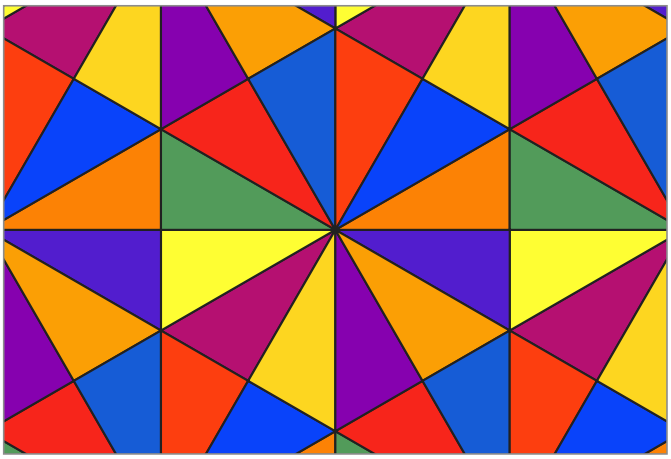
Uniform colouring 111112

Colouring: 111112 · tetrad (rectangle) (violet)



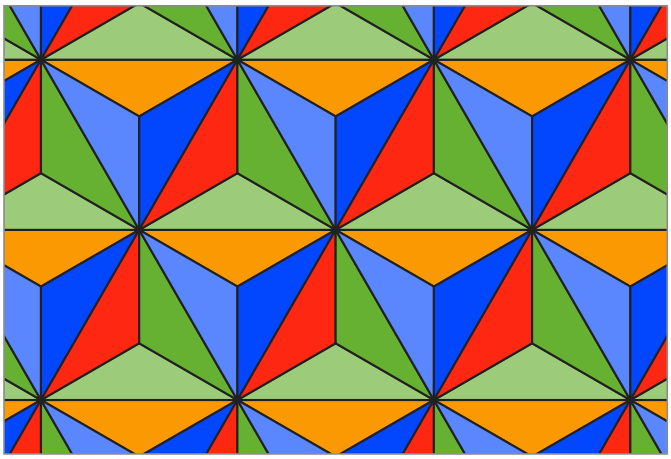
Uniform colouring 121314

Colouring: 121314 · square (red)



Kisrhombille: 30-60-90 triangles (dual of 4.6.12)

Colouring: by orientation · rainbow (yellow)



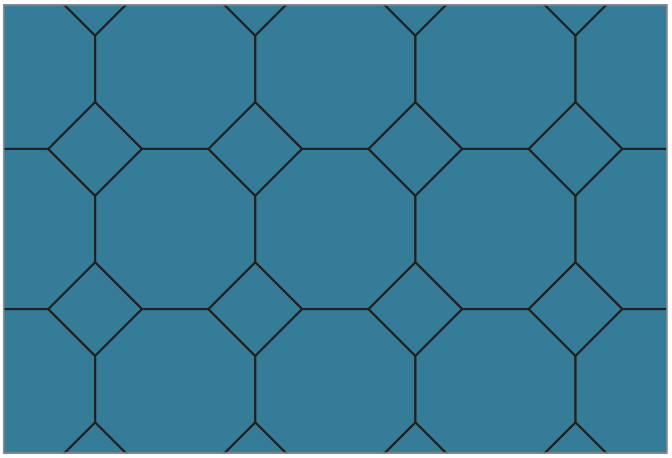
Kisdeltile: 30-30-120 triangles (dual of 3.12.12)

Colouring: by orientation · tetrad (rectangle) (blue)

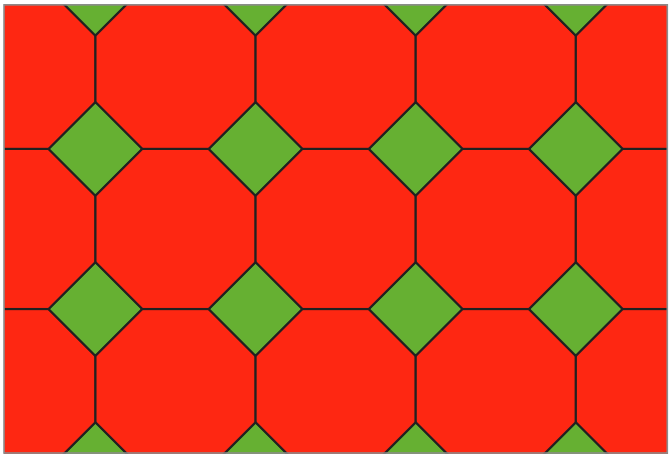
Truncated square tiling

Wikipedia: Truncated square tiling — en.wikipedia.org/wiki/Truncated_square_tiling

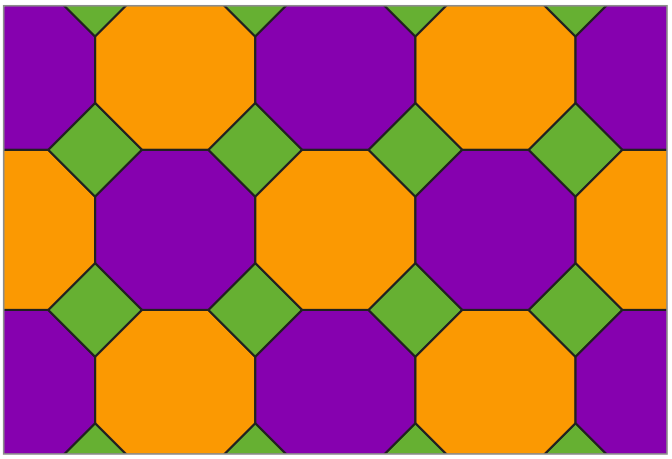
One square and two regular octagons at each vertex (4.8.8), the floor of many kitchens. The squares can take two sizes, and in the limit half of them vanish (chamfered tiling). Its dual is the tetrakis square tiling.



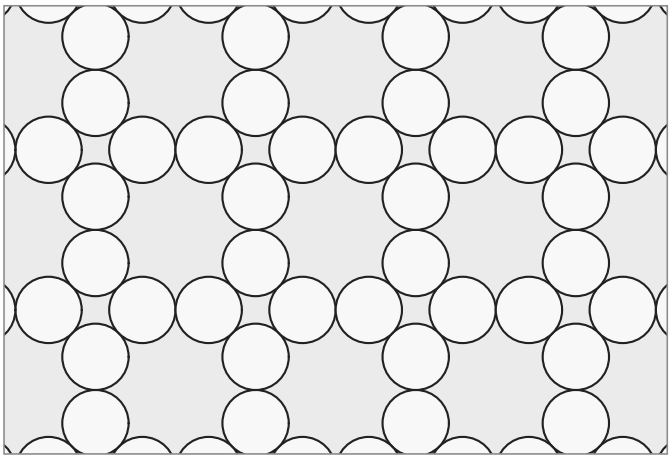
Truncated square tiling 4.8.8
Colouring: one colour · monochromatic (blue-green)



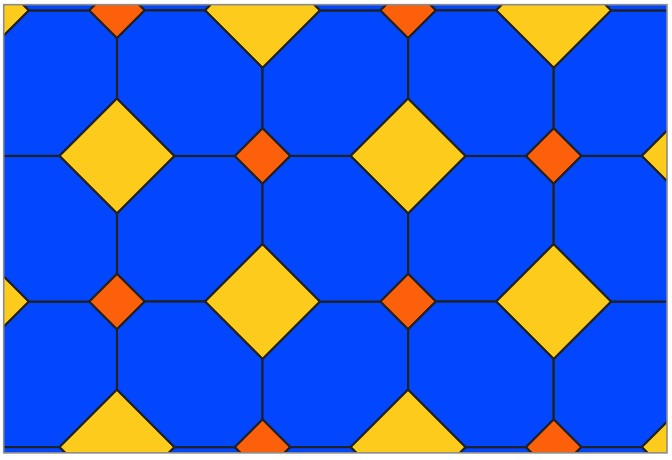
Uniform colouring 122
Colouring: 122 · complementary (red)



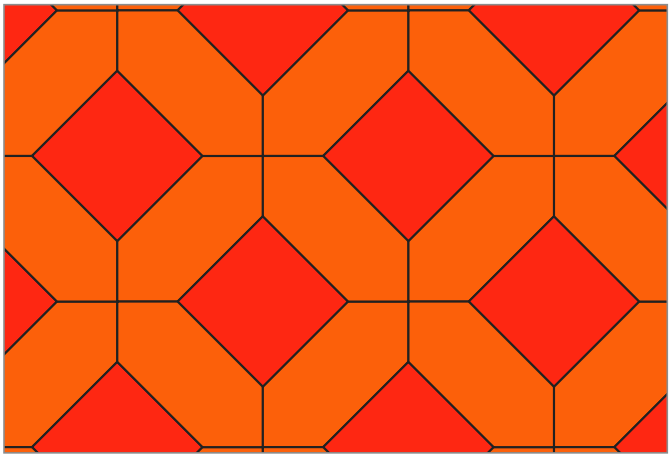
Uniform colouring 123
Colouring: 123 · triad (orange)



Circle packing: a circle at every vertex
Colouring: one colour · greys



Squares of alternating sizes
Colouring: by kind of tile · split complementary (blue)

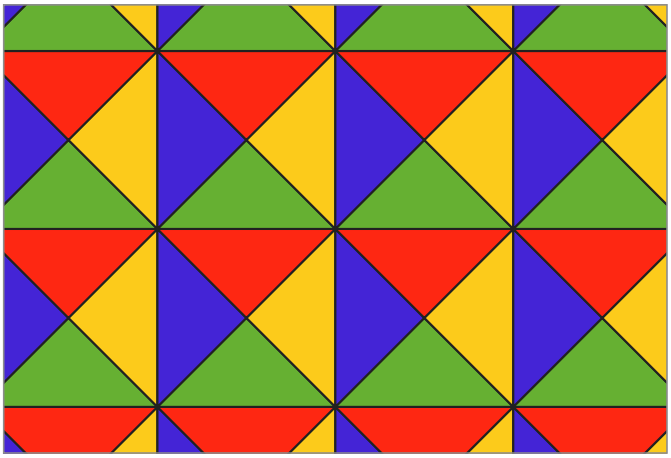


Chamfered square tiling (small squares vanish)
Colouring: by kind of tile · analogous (red-orange)

Truncated square tiling (continued)

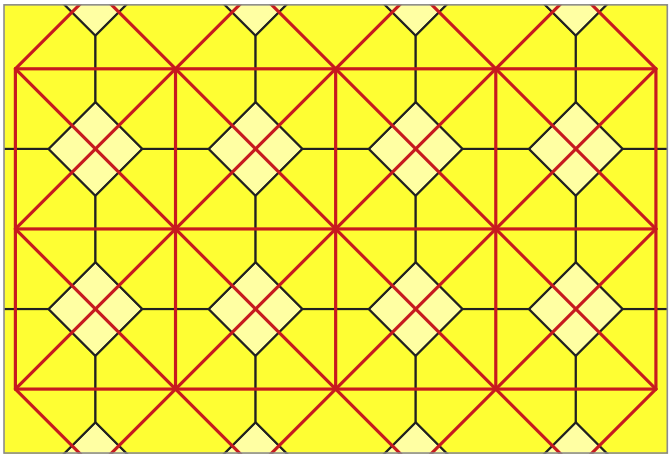
Wikipedia: Truncated square tiling — en.wikipedia.org/wiki/Truncated_square_tiling

One square and two regular octagons at each vertex (4.8.8), the floor of many kitchens. The squares can take two sizes, and in the limit half of them vanish (chamfered tiling). Its dual is the tetrakis square tiling.



Tetrakis square tiling (dual of 4.8.8)

Colouring: by orientation · square (yellow-orange)



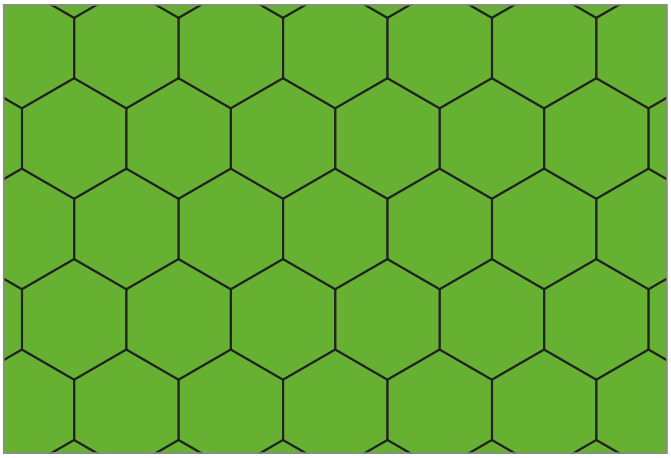
With its dual drawn over it

Colouring: 122 · monochromatic (yellow)

Hexagonal tiling

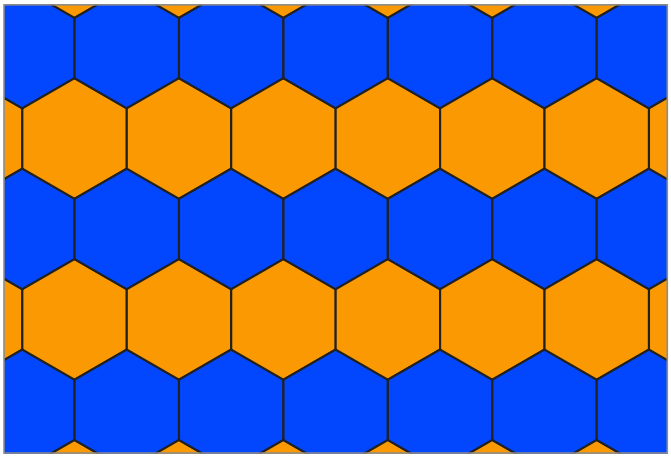
Wikipedia: Hexagonal tiling — en.wikipedia.org/wiki/Hexagonal_tiling

Three hexagons at each vertex, as in a honeycomb. Hexagons can be coloured with 2, 3, 4 or 7 colours in a regular way; with 7 every hexagon touches all the other colours. Cutting some hexagons into triangles gives 2-uniform tilings.



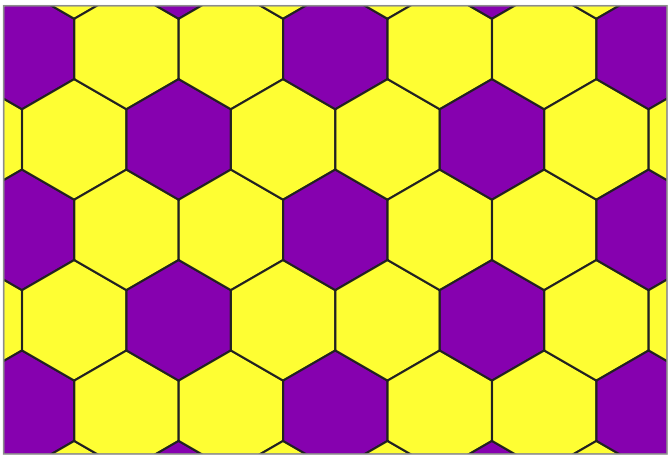
Regular hexagonal tiling {6,3}

Colouring: 1 · monochromatic (green)



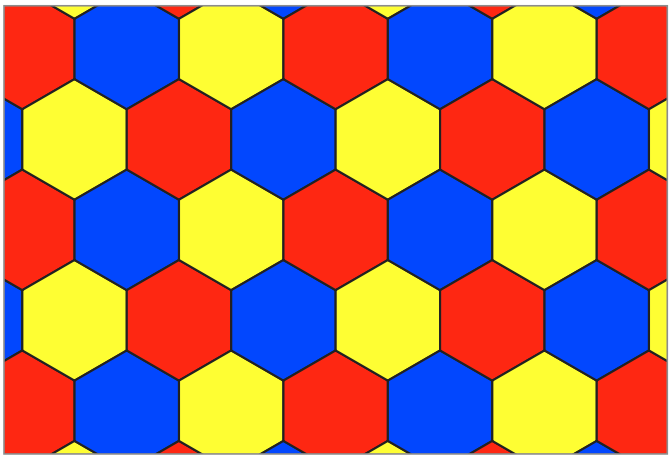
Two colours, in rows

Colouring: 2 · complementary (blue)



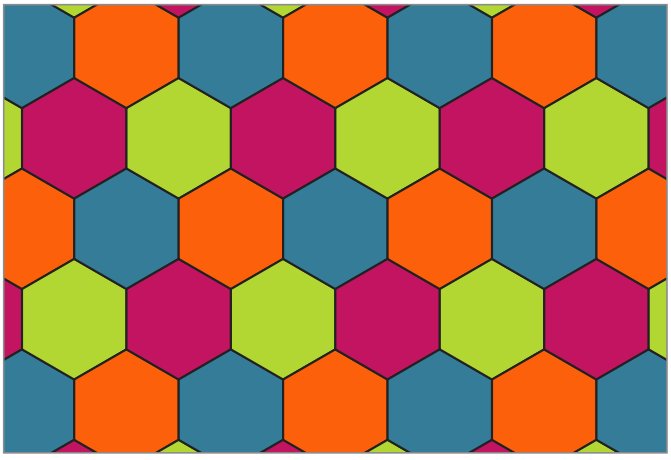
Two colours, one hexagon in three

Colouring: 2-1 · complementary (yellow)



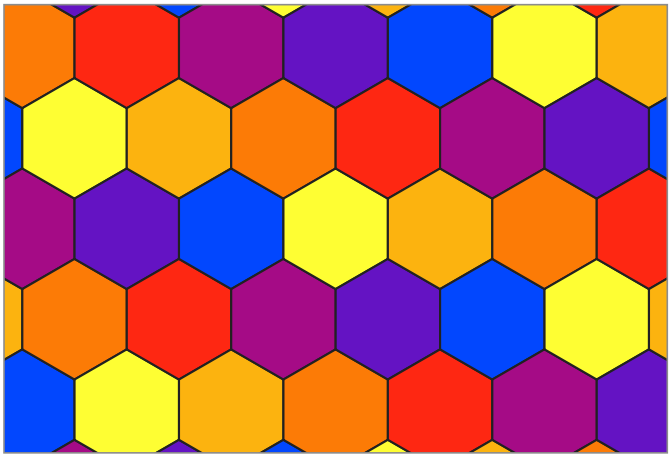
Three colours: neighbours always differ

Colouring: 3 · triad (red)



Four colours

Colouring: 4 · tetrad (rectangle) (blue-green)



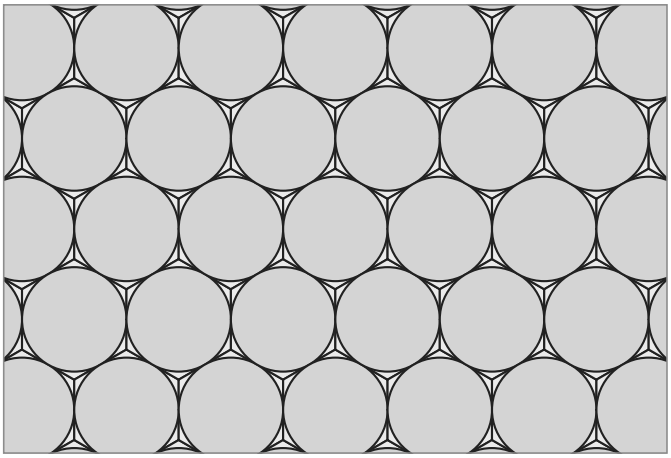
Seven colours: each hexagon touches all the others

Colouring: 7 · rainbow (yellow)

Hexagonal tiling (continued)

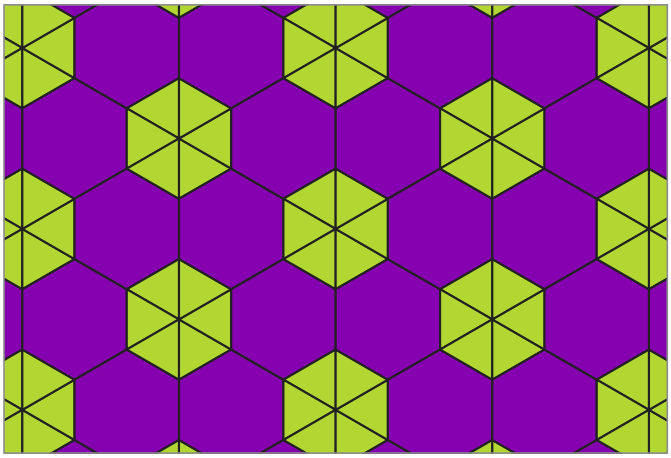
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Three hexagons at each vertex, as in a honeycomb. Hexagons can be coloured with 2, 3, 4 or 7 colours in a regular way; with 7 every hexagon touches all the other colours. Cutting some hexagons into triangles gives 2-uniform tilings.



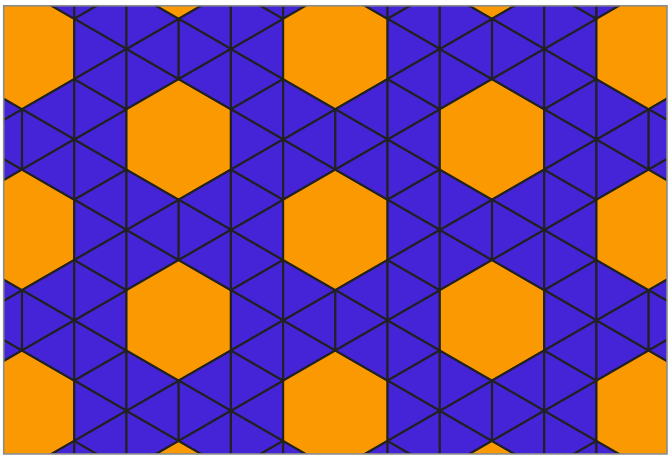
Densest packing of circles

Colouring: 1 · greys



2-uniform: a third of the hexagons cut into triangles

Colouring: by kind of tile · split complementary (violet)



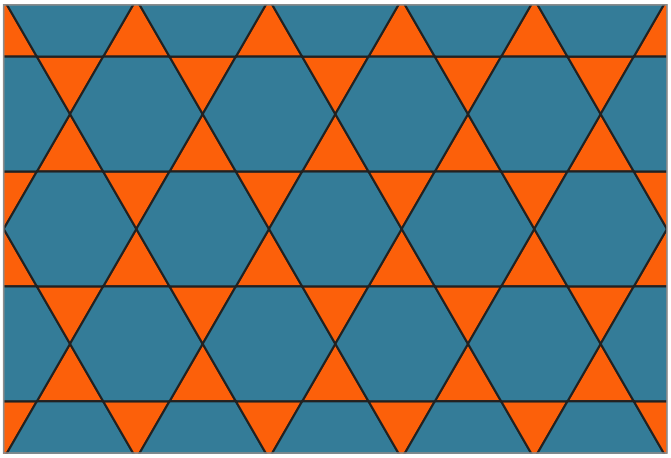
2-uniform: two thirds of the hexagons cut into triangles

Colouring: by kind of tile · split complementary (orange)

Trihexagonal tiling

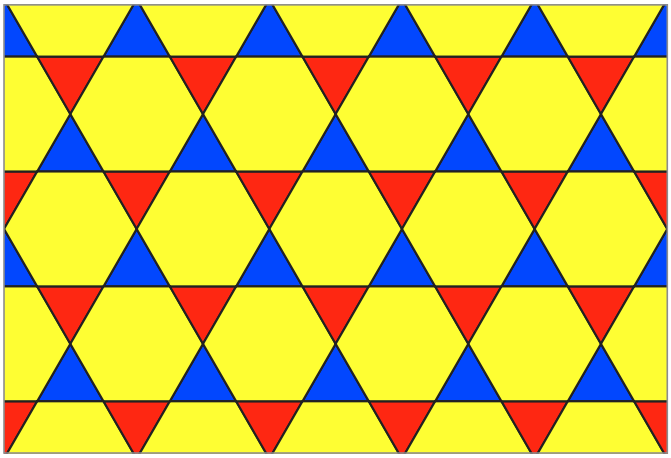
Wikipedia: Trihexagonal tiling — en.wikipedia.org/wiki/Trihexagonal_tiling

Triangles and hexagons alternate around each vertex (3.6.3.6); its edges are three families of straight lines, the kagome pattern of Japanese baskets. Turning or resizing the triangles keeps the same structure.



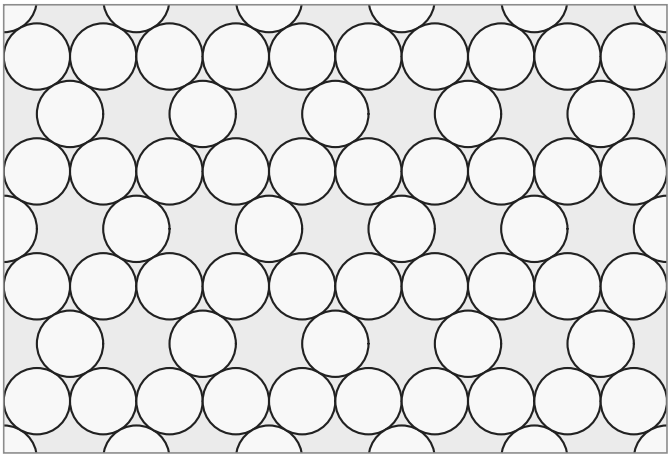
Uniform colouring 1212

Colouring: 1212 · complementary (blue-green)



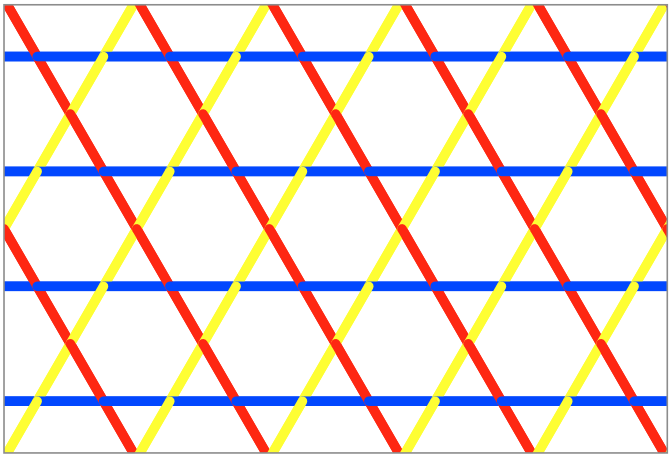
Uniform colouring 1232

Colouring: 1232 · triad (yellow)



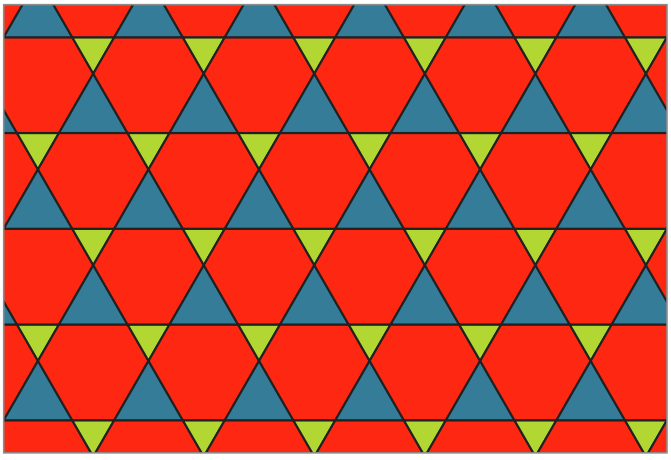
Circle packing: a circle at every vertex

Colouring: one colour · greys



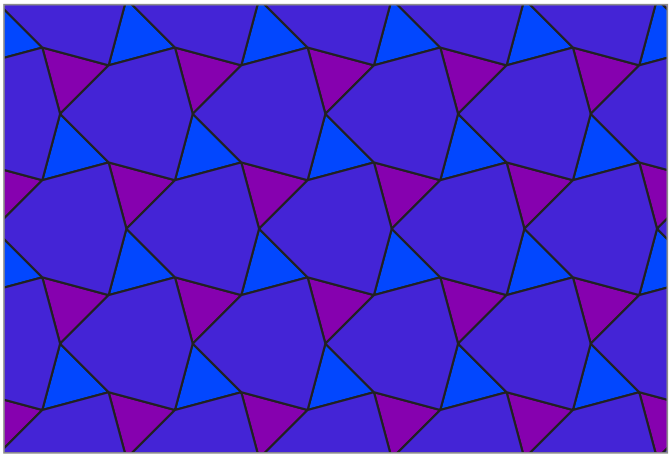
Kagome: edges coloured by direction

Colouring: one colour · triad (blue)



Triangles of two sizes

Colouring: by kind of tile · split complementary (red)



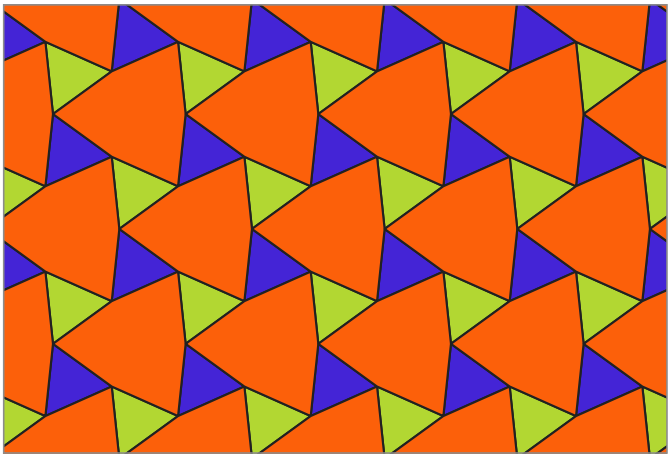
Turned triangles: the hexagons become stars

Colouring: by kind of tile · analogous (blue-violet)

Trihexagonal tiling (continued)

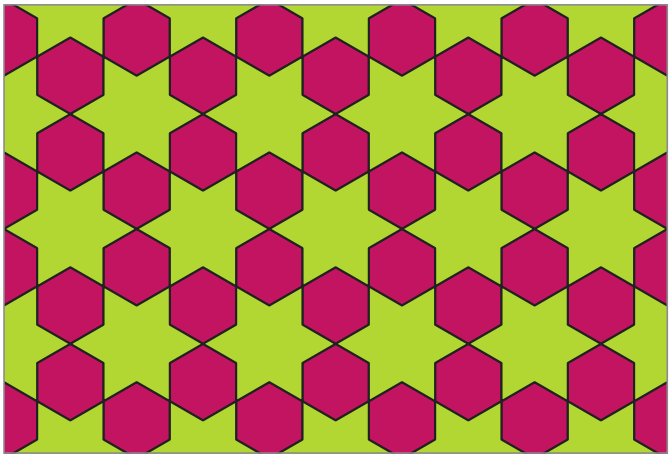
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Turned further

Colouring: by kind of tile · triad (red-orange)



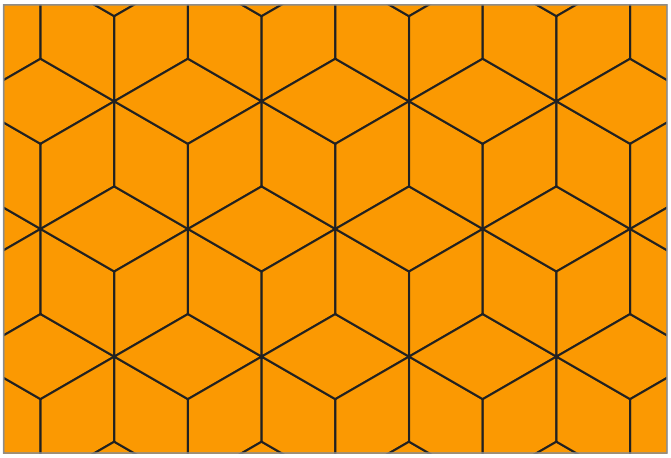
Hexagons and six-pointed stars

Colouring: by kind of tile · complementary (red-violet)

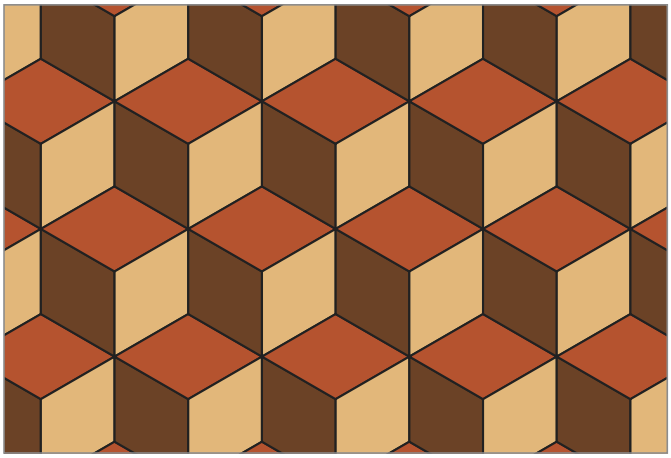
Rhombille tiling

Wikipedia: Rhombille tiling — en.wikipedia.org/wiki/Rhombille_tiling

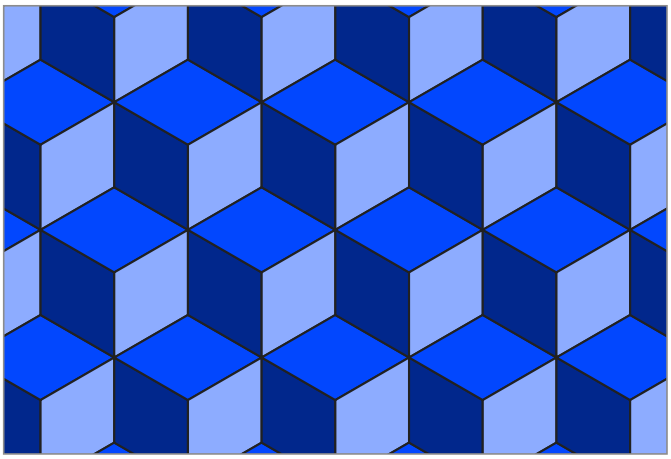
Identical 60° rhombi, three around one corner and six around the other: the dual of the trihexagonal tiling. With three values of one hue it reads as a stack of cubes, a pattern found on old floors such as Siena Cathedral.



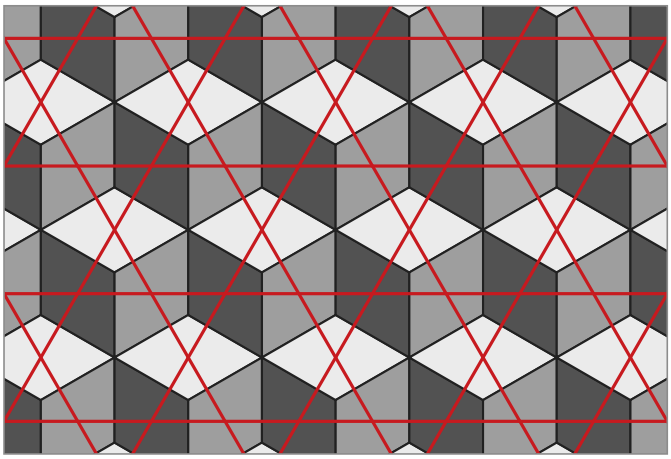
Rhombille tiling: 60° rhombi
Colouring: one colour · monochromatic (orange)



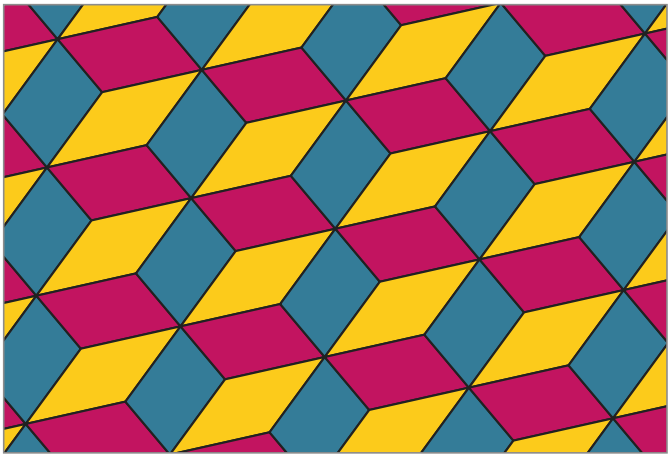
Three shades by orientation, as in Siena Cathedral
Colouring: by orientation · earth tones



Stacked cubes: one hue in three values
Colouring: by orientation · monochromatic (blue)



Over its dual, the trihexagonal tiling
Colouring: by orientation · greys

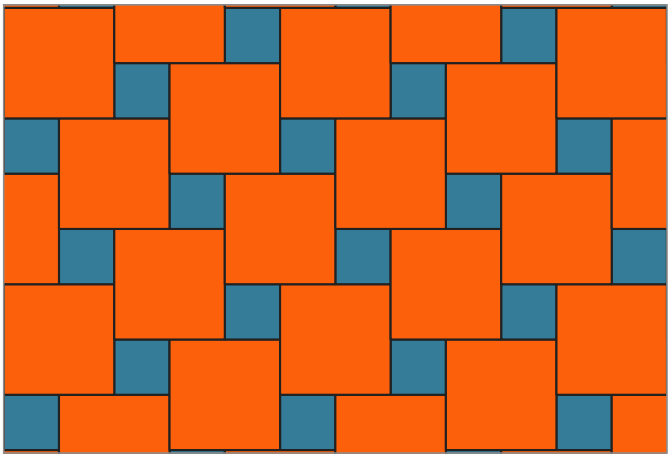


Skewed and turned: parallelograms
Colouring: by orientation · triad (yellow-orange)

Pythagorean tiling

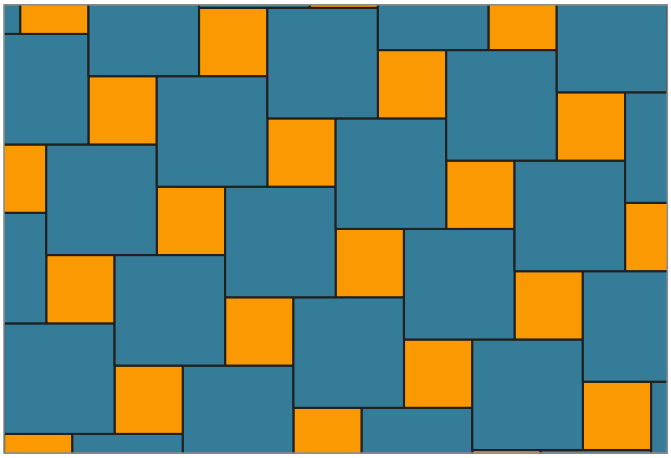
Wikipedia: Pythagorean tiling — en.wikipedia.org/wiki/Pythagorean_tiling

Squares of two sizes, each touching four squares of the other size. It is the picture behind a classic proof of Pythagoras' theorem, and a common floor pattern (pinwheel or hopscotch).



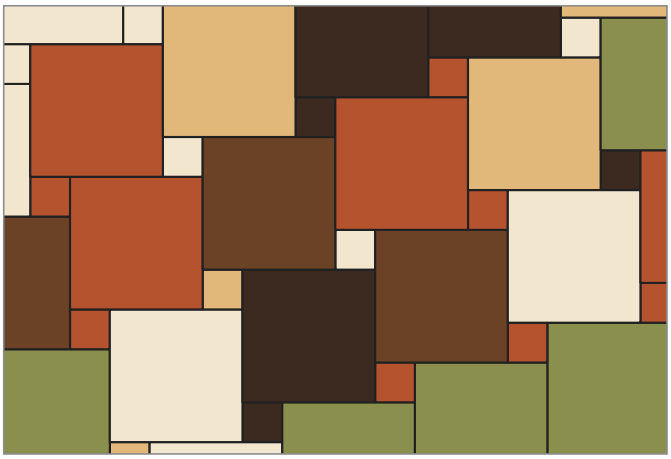
Squares of sides 1 and 1/2

Colouring: by kind of tile · complementary (red-orange)



Sides in the golden ratio

Colouring: by kind of tile · split complementary (blue-green)



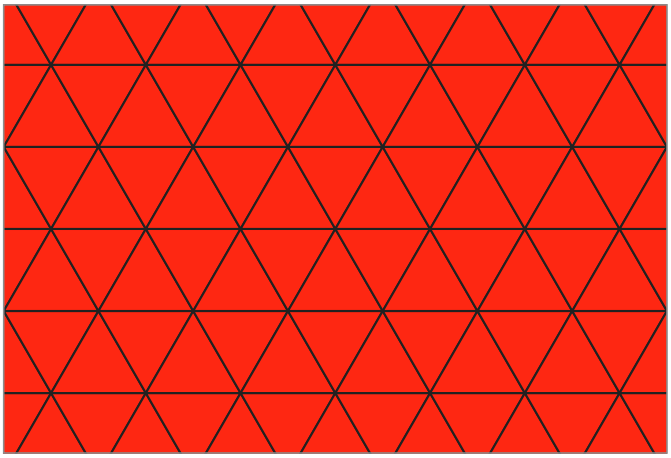
Sides 1 and 0.3, random earth tones

Colouring: random · earth tones

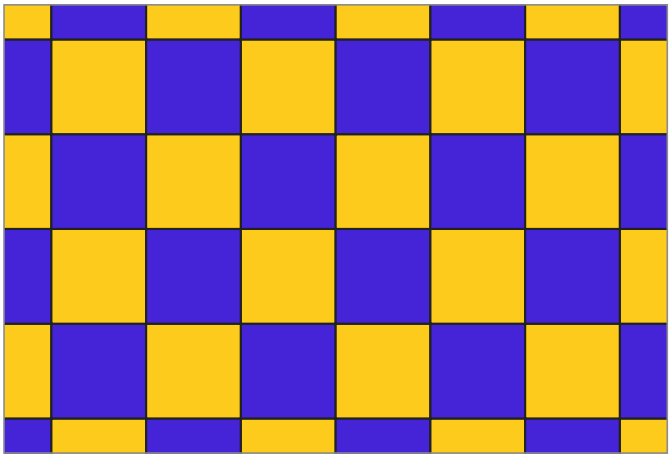
Regular polygons

Wikipedia: Euclidean tilings by convex regular polygons — en.wikipedia.org/wiki/Euclidean_tilings_by_convex_regular_polygons

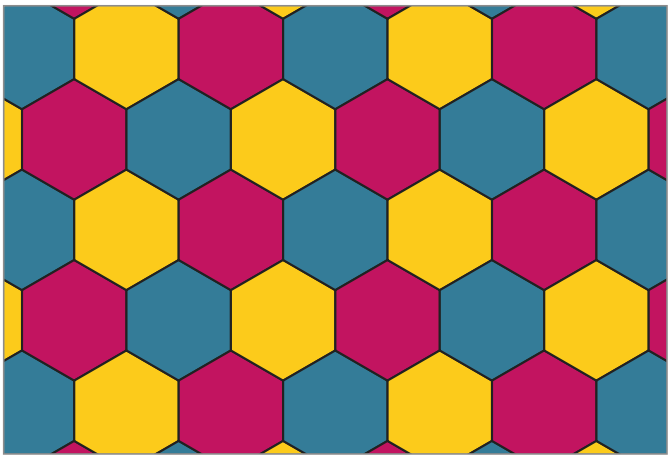
The eleven uniform (Archimedean) tilings by regular polygons: every vertex has the same polygons around it, written as their numbers of sides (3.4.6.4 is triangle, square, hexagon, square). Mixing vertex types gives k-uniform tilings.



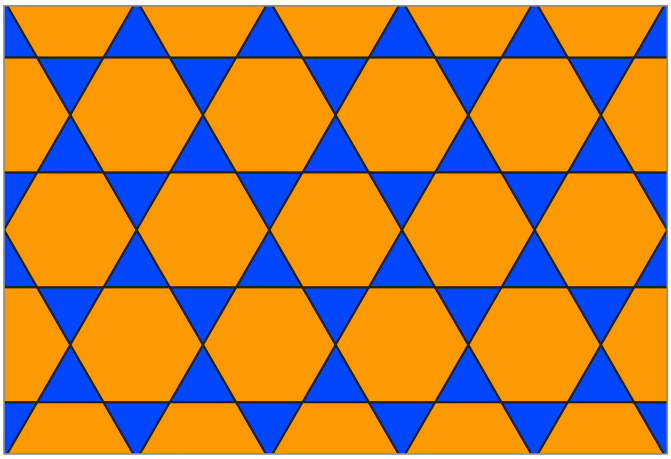
3.3.3.3.3.3 triangular
Colouring: 111111 · monochromatic (red)



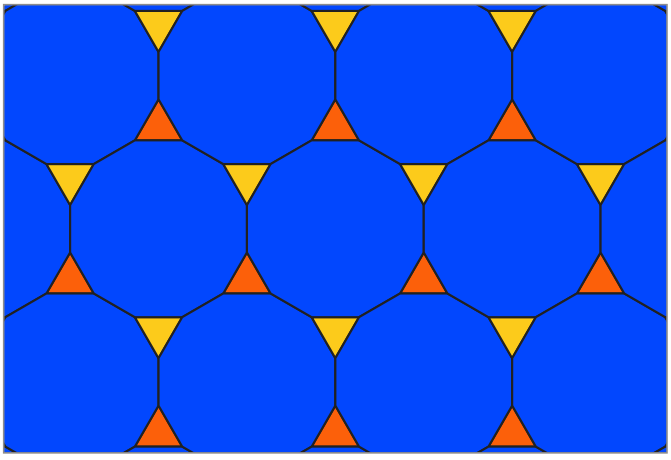
4.4.4.4 square
Colouring: checkerboard · complementary (yellow-orange)



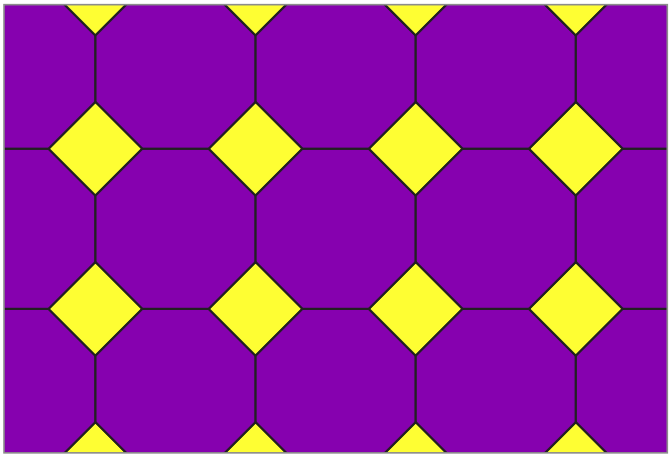
6.6.6 hexagonal
Colouring: 3 · triad (blue-green)



3.6.3.6 trihexagonal
Colouring: 1212 · complementary (orange)



3.12.12 truncated hexagonal
Colouring: by kind of tile · split complementary (blue)

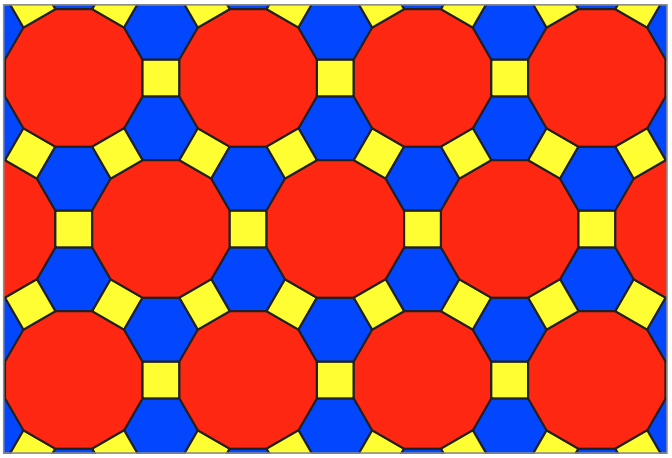


4.8.8 truncated square
Colouring: by kind of tile · complementary (violet)

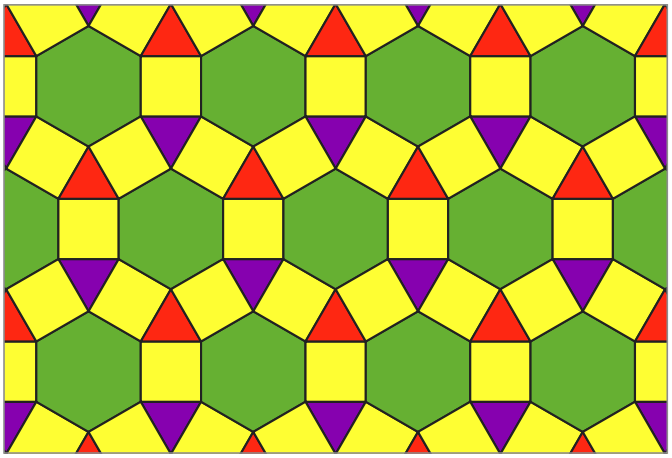
Regular polygons (continued)

Wikipedia: Euclidean tilings by convex regular polygons — en.wikipedia.org/wiki/Euclidean_tilings_by_convex_regular_polygons

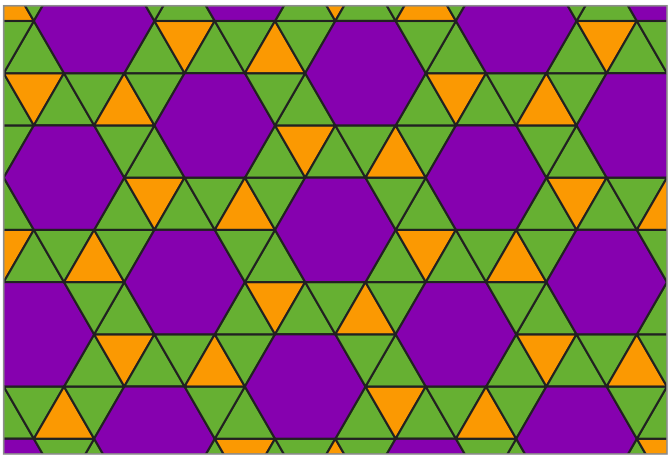
The eleven uniform (Archimedean) tilings by regular polygons: every vertex has the same polygons around it, written as their numbers of sides (3.4.6.4 is triangle, square, hexagon, square). Mixing vertex types gives k-uniform tilings.



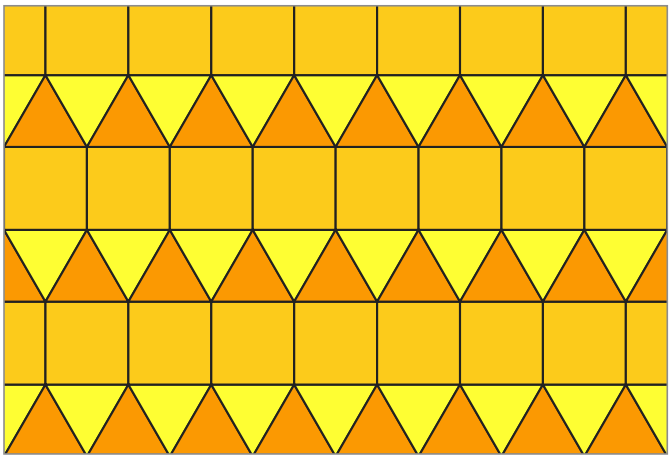
4.6.12 truncated trihexagonal
Colouring: by kind of tile · triad (red)



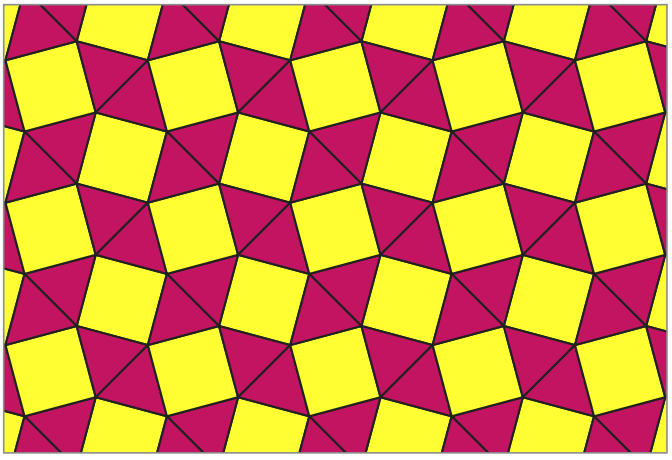
3.4.6.4 rhombitrihexagonal
Colouring: by kind of tile · tetrad (rectangle) (green)



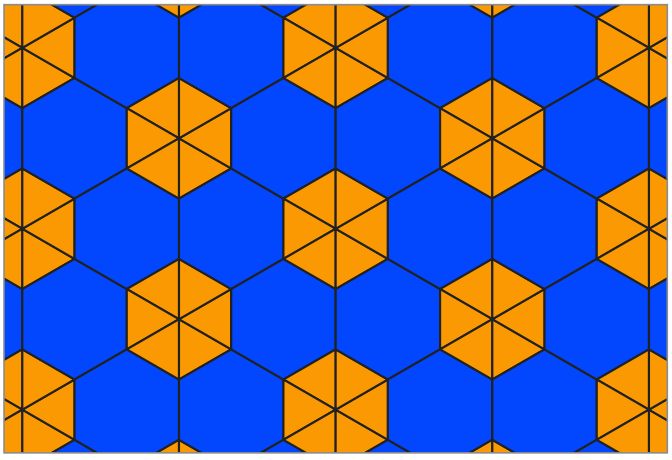
3.3.3.3.6 snub hexagonal
Colouring: by kind of tile · triad (violet)



3.3.3.4.4 elongated triangular
Colouring: by kind of tile · analogous (yellow-orange)



3.3.4.3.4 snub square
Colouring: by kind of tile · split complementary (yellow)

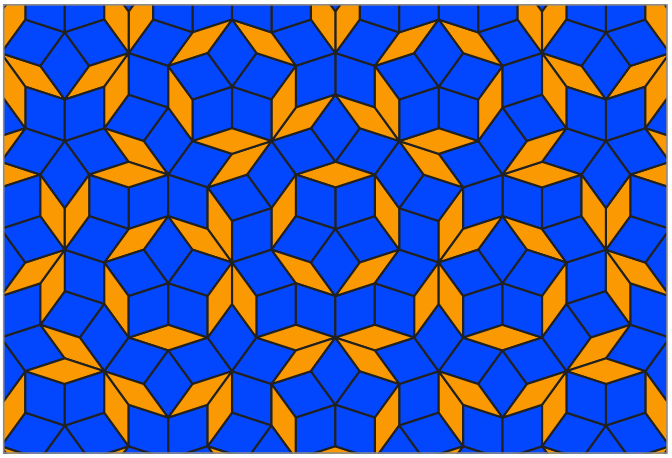


2-uniform: a third of the hexagons cut into triangles
Colouring: by kind of tile · complementary (blue)

Penrose tilings

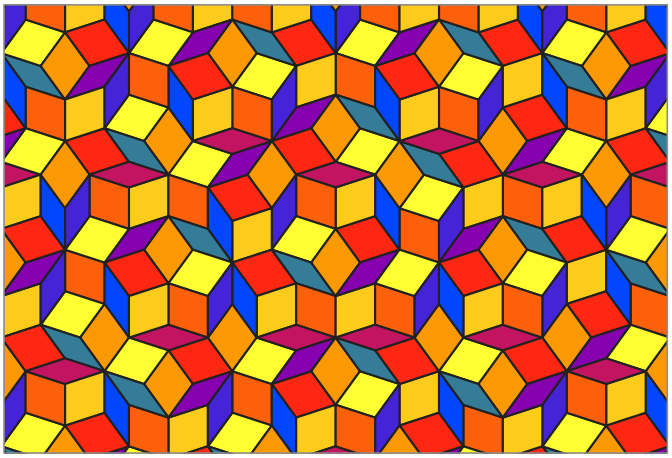
Wikipedia: Penrose tiling — en.wikipedia.org/wiki/Penrose_tiling

Two tiles that cover the plane only without repeating: thick and thin rhombs (P3), or kites and darts (P2). They are drawn by repeatedly cutting golden triangles (deflation) and show five-fold symmetry everywhere.



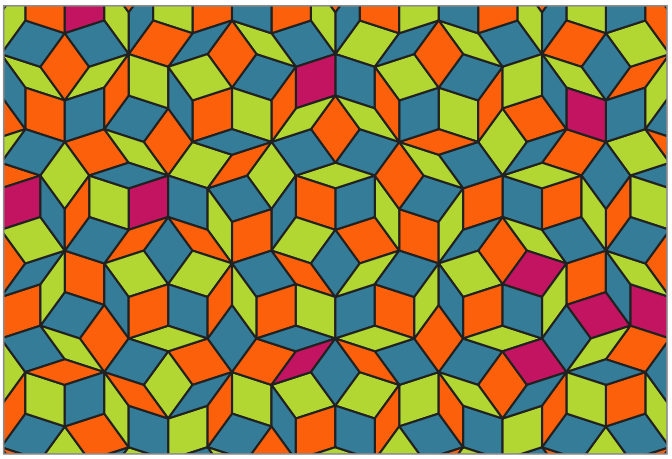
P3: thick and thin rhombs

Colouring: by kind of tile · complementary (blue)



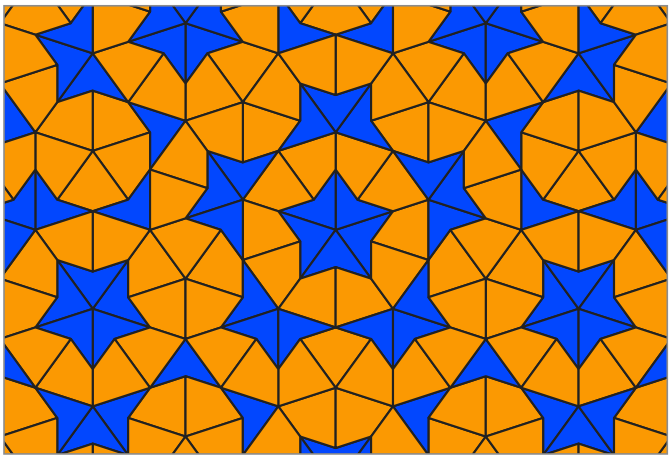
P3 by the ten orientations of the rhombs

Colouring: by orientation · rainbow (yellow)



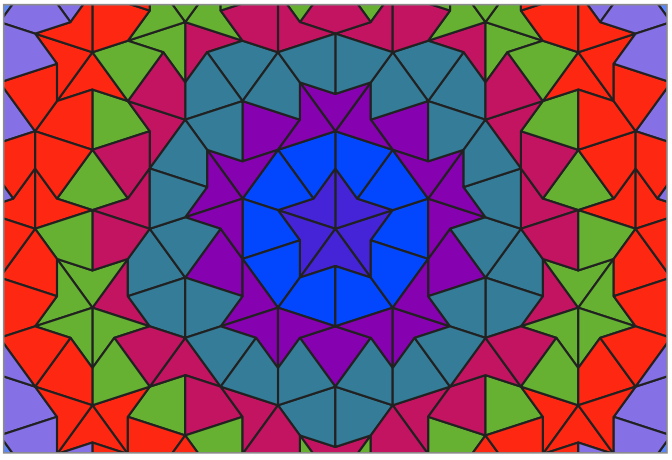
P3, neighbouring rhombs differ

Colouring: neighbours differ · tetrad (rectangle) (blue-green)



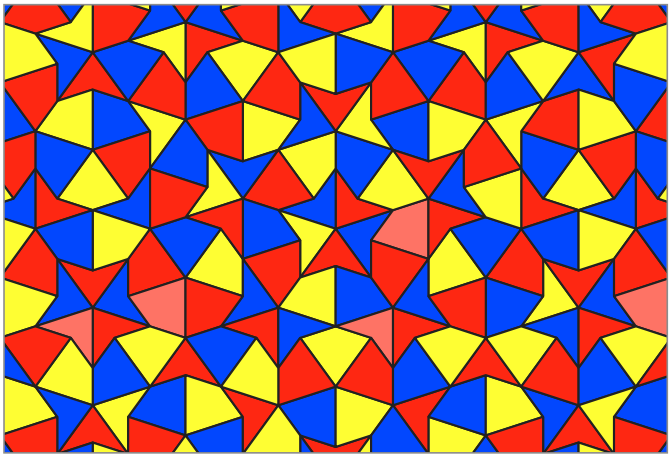
P2: kites and darts

Colouring: by kind of tile · complementary (orange)



P2 in rings round the centre

Colouring: rings from the centre · analogous (blue-violet)



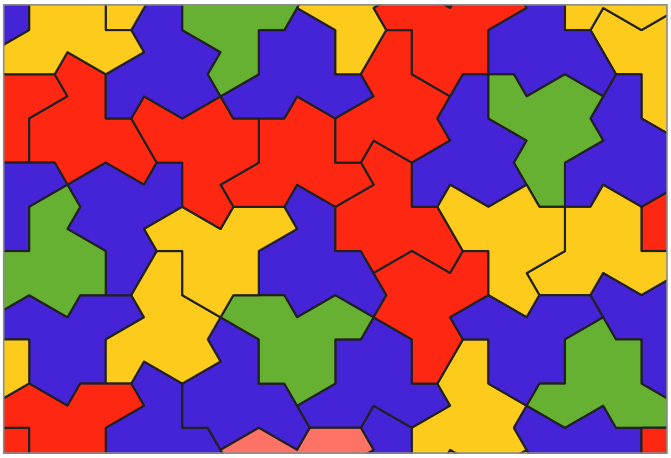
P2, neighbouring tiles differ

Colouring: neighbours differ · triad (red)

Aperiodic tilings

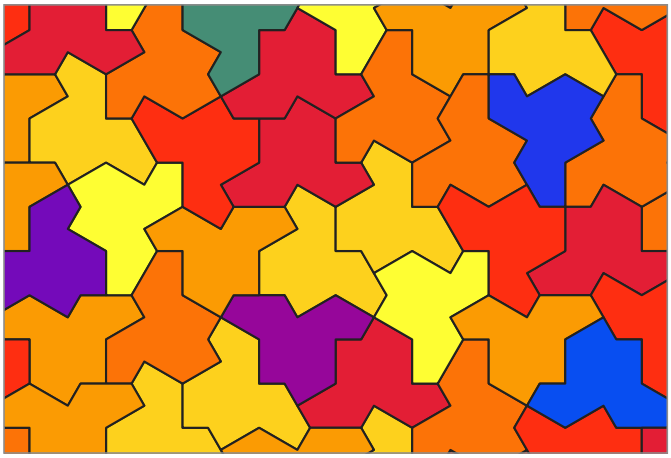
Wikipedia: Aperiodic tiling — en.wikipedia.org/wiki/Aperiodic_tiling

Tilings that never repeat. The hat (2023) is a single shape, used with its mirror image, that tiles only aperiodically; Robinson's six square tiles force nested squares of ever larger sizes.



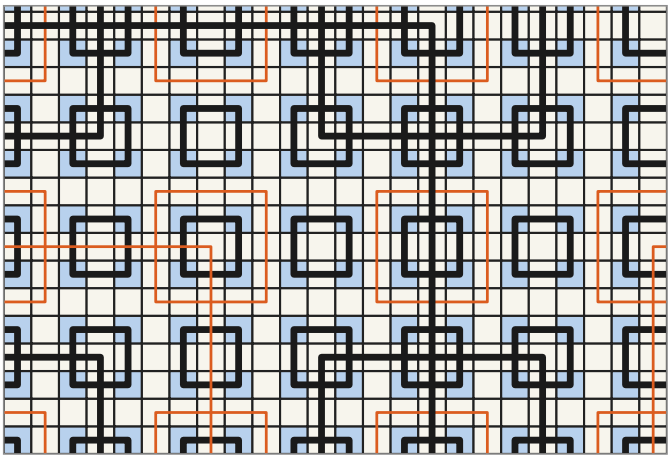
The hat: one shape and its mirror image

Colouring: by kind of tile · square (red)



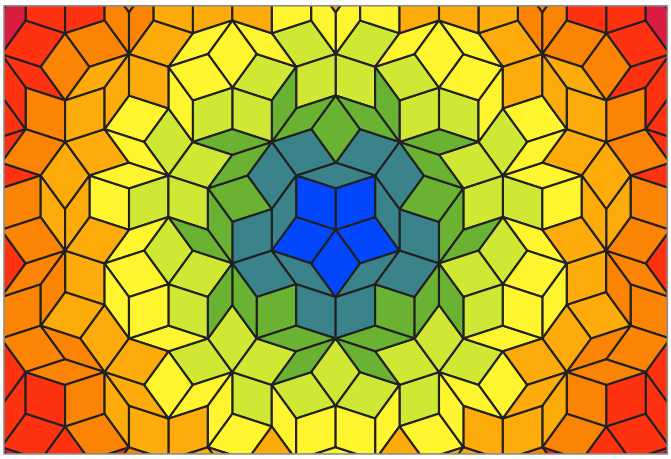
Hats by orientation

Colouring: by orientation · rainbow (yellow)



Robinson tiles: their marks draw nested squares

Colouring: by kind of tile



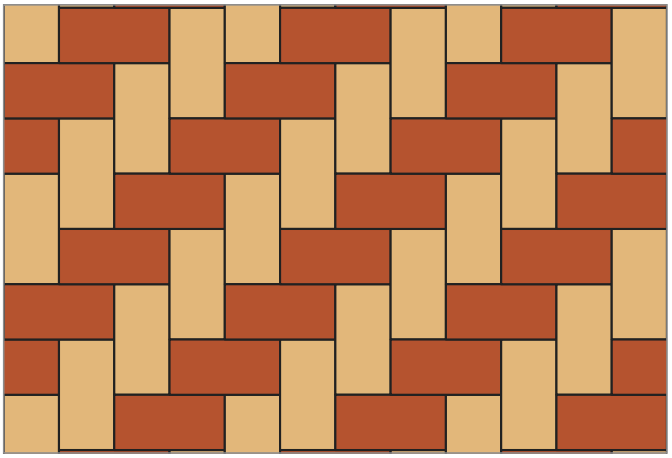
P3 in rings round the centre

Colouring: rings from the centre · rainbow (blue)

Herringbone pattern

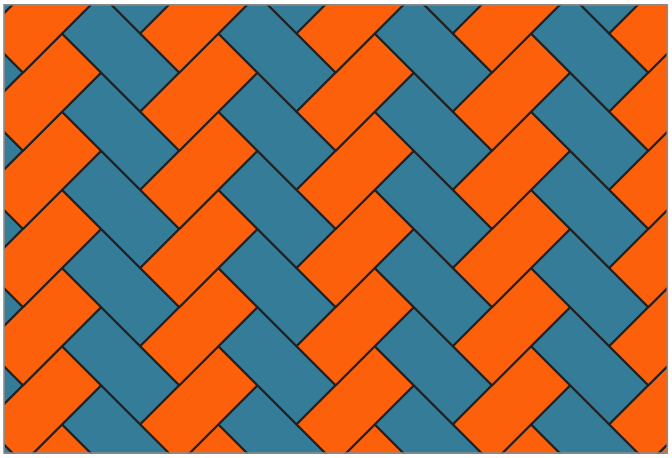
Wikipedia: Herringbone pattern — en.wikipedia.org/wiki/Herringbone_pattern

Rectangles laid in a staircase, each at right angles to its neighbours: the classic brick and parquet pattern. Longer bricks, a 45° turn or parallelograms change its look; the basket weave pairs the bricks instead.



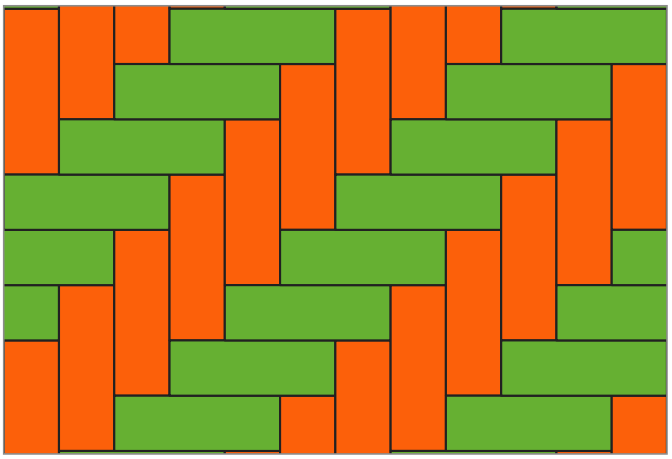
Bricks 2:1

Colouring: by kind of tile · earth tones



Turned 45°

Colouring: by kind of tile · complementary (red-orange)



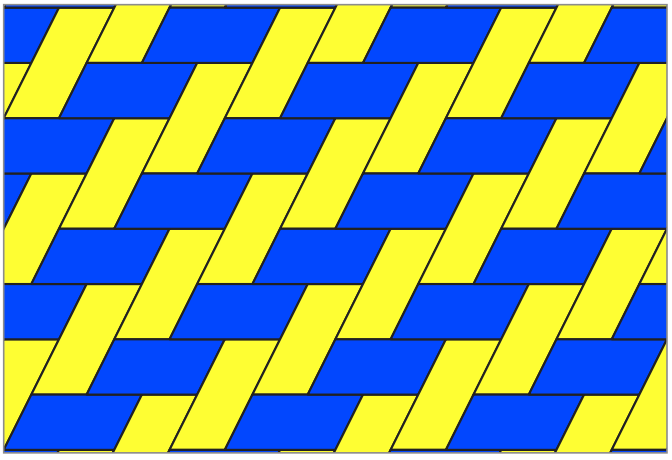
Bricks 3:1

Colouring: by kind of tile · split complementary (green)



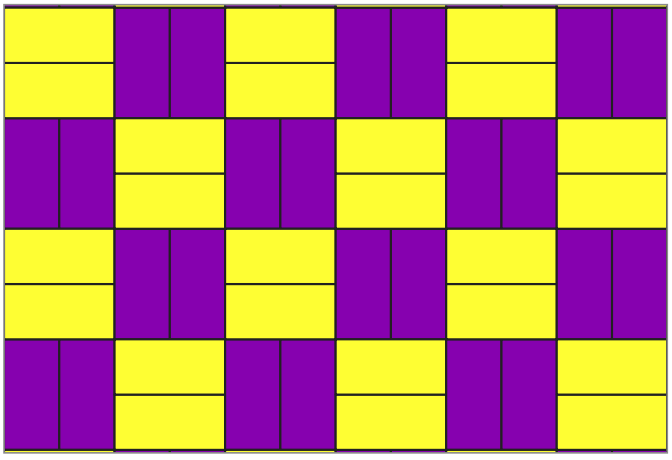
Long boards, about 6.6:1

Colouring: random · earth tones



Parallelograms

Colouring: by kind of tile · triad (blue)



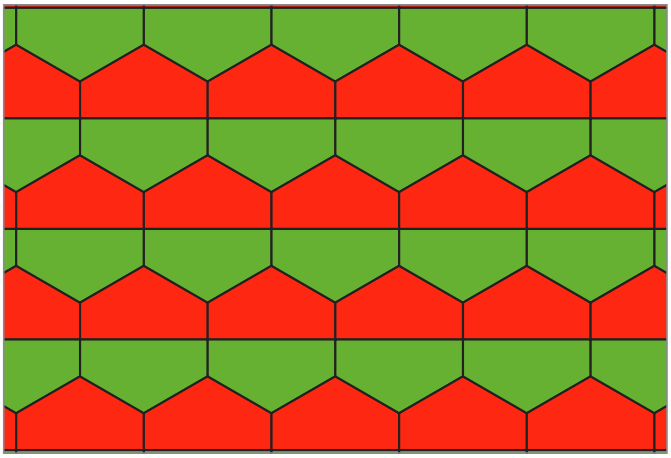
Basket weave

Colouring: by kind of tile · complementary (yellow)

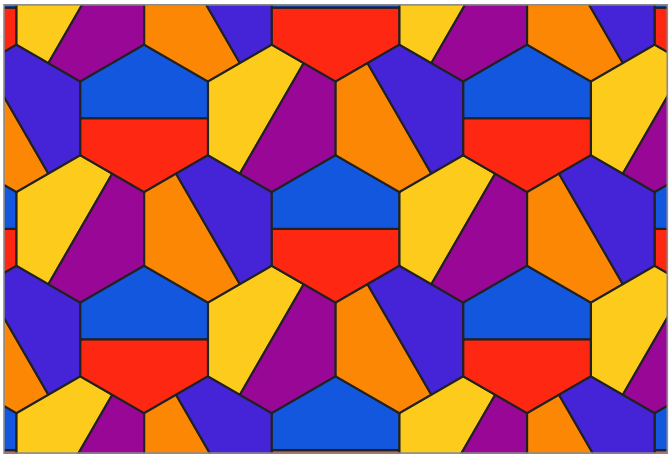
Pentagonal tilings

Wikipedia: Pentagonal tiling — en.wikipedia.org/wiki/Pentagonal_tiling

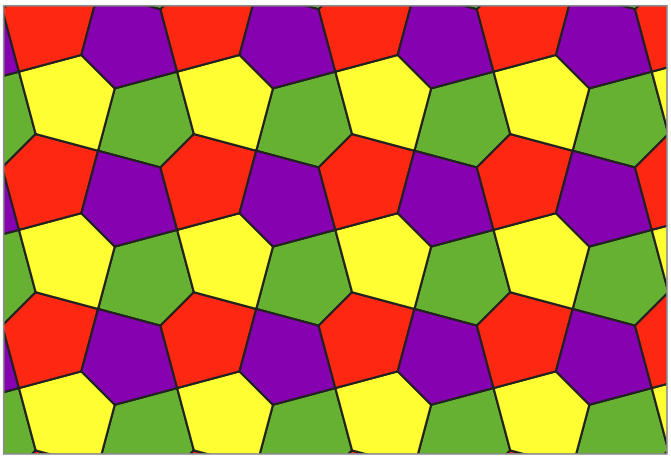
Regular pentagons cannot tile the plane, but 15 families of convex pentagons can. Halving hexagons gives type 1; the duals of three uniform tilings give the Cairo, prismatic and floret pentagonal tilings.



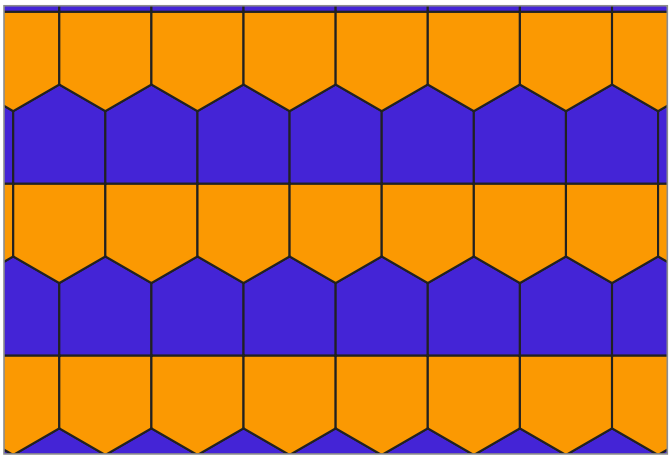
Type 1: hexagons cut in half
Colouring: by kind of tile · complementary (green)



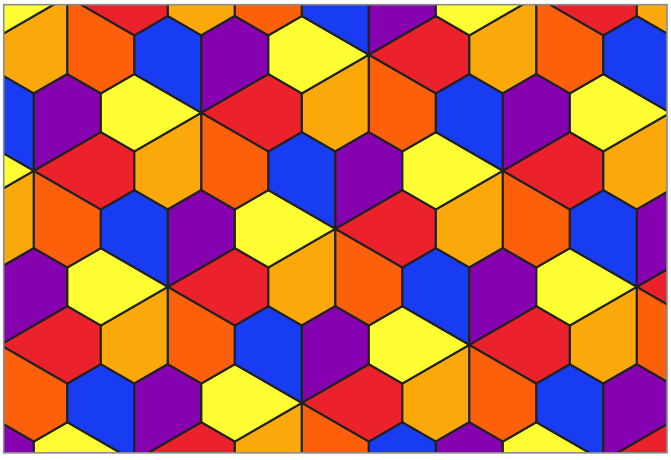
Type 1, cut in three directions
Colouring: by orientation · rainbow (yellow-orange)



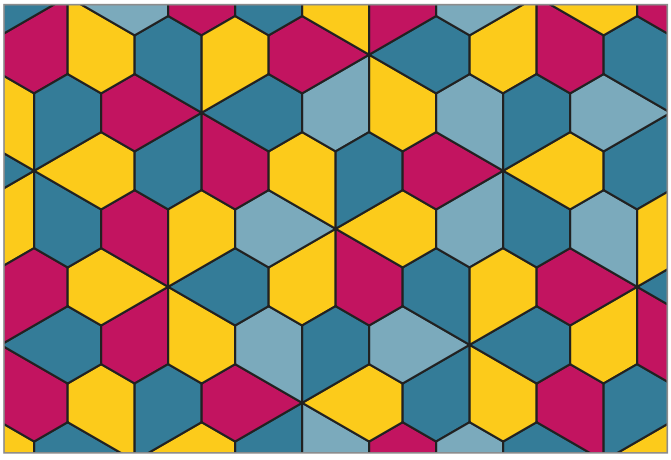
Cairo tiling (types 2 and 4)
Colouring: by orientation · tetrad (rectangle) (red)



Prismatic pentagonal (dual of 3.3.3.4.4)
Colouring: by orientation · split complementary (orange)



Floret pentagonal (dual of 3.3.3.3.6)
Colouring: by orientation · rainbow (yellow)

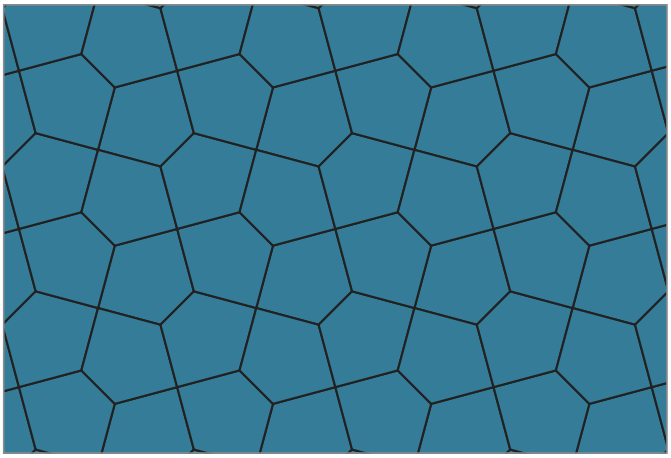


Floret pentagonal, neighbours differ
Colouring: neighbours differ · triad (blue-green)

Cairo pentagonal tiling

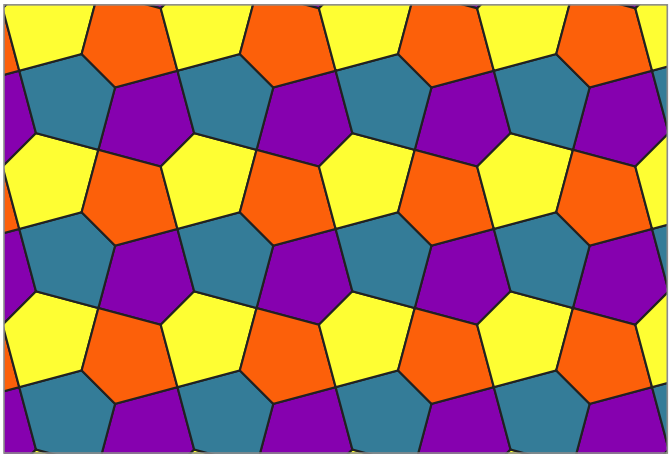
Wikipedia: Cairo pentagonal tiling — en.wikipedia.org/wiki/Cairo_pentagonal_tiling

Pentagons with four long sides and one short side, the dual of the snub square tiling, named after the streets of Cairo. Four pentagons meet at some vertices, three at others, and pairs of pentagons form hexagons.



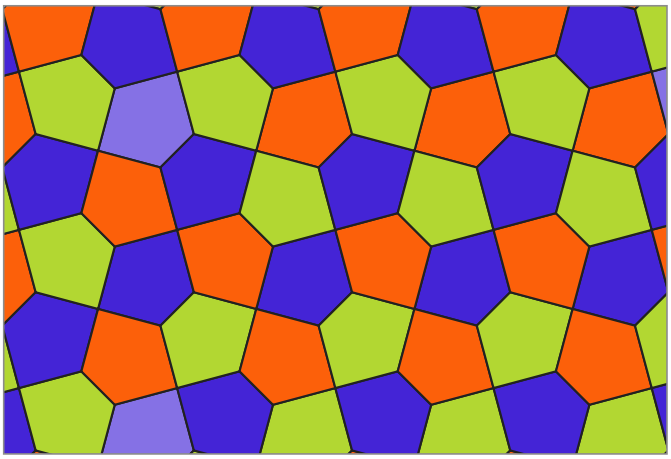
Cairo tiling: dual of the snub square tiling

Colouring: one colour · monochromatic (blue-green)



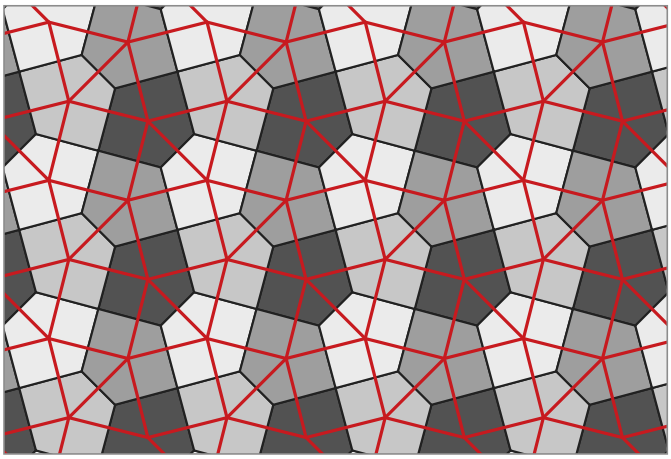
Four orientations

Colouring: by orientation · square (yellow)



Neighbouring pentagons differ

Colouring: neighbours differ · triad (blue-violet)



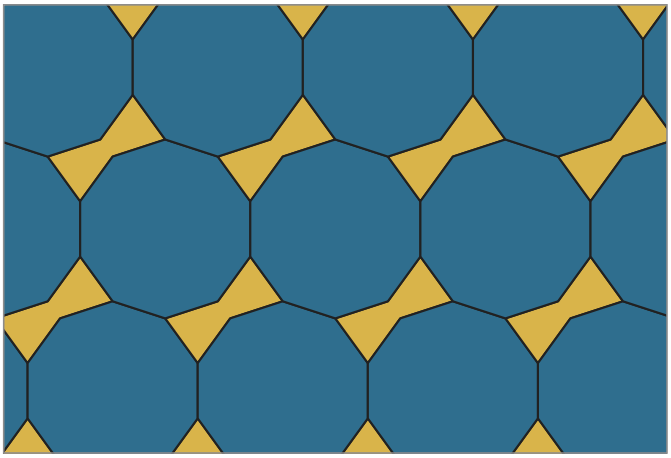
Over its dual, the snub square tiling

Colouring: by orientation · greys

Girih tiles

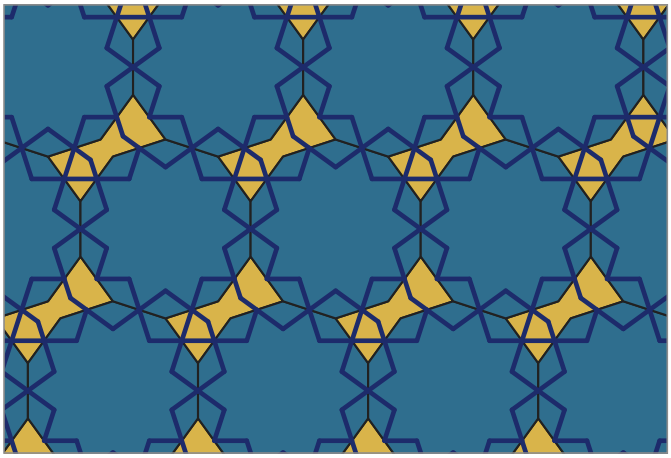
Wikipedia: Girih tile — en.wikipedia.org/wiki/Girih_tile

Five equal-sided tiles (decagon, pentagon, hexagon, bowtie, rhombus) used in Persian architecture. Lines drawn on them cross every edge at its midpoint at 54°, so they join across tiles into the visible star pattern.



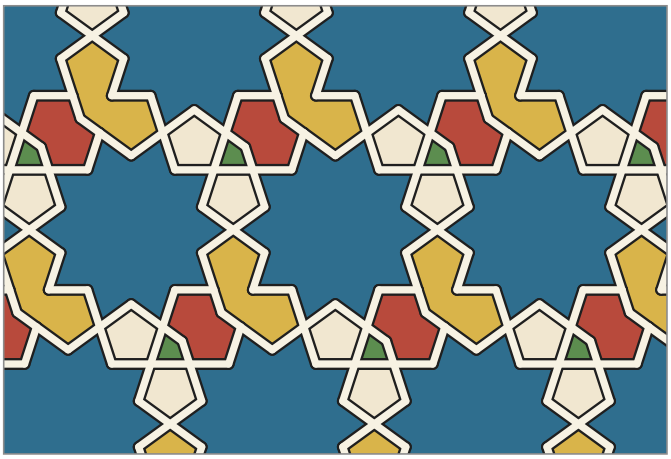
Decagons and bowties

Colouring: by kind of tile · girih palette



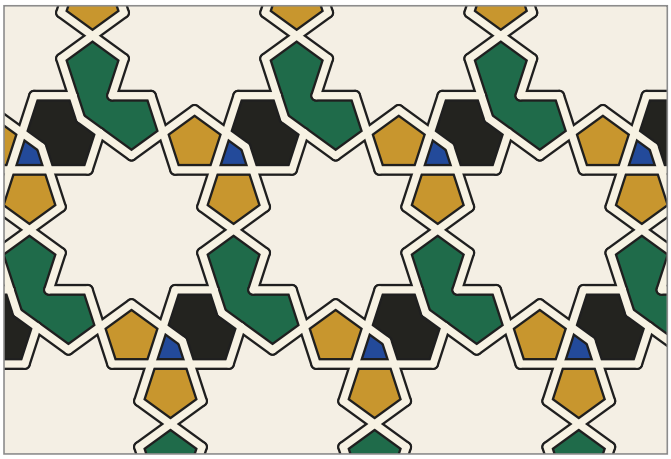
Girih lines, crossing every edge at 54°

Colouring: by kind of tile · girih palette



The girih pattern, coloured

Colouring: by shape · girih palette



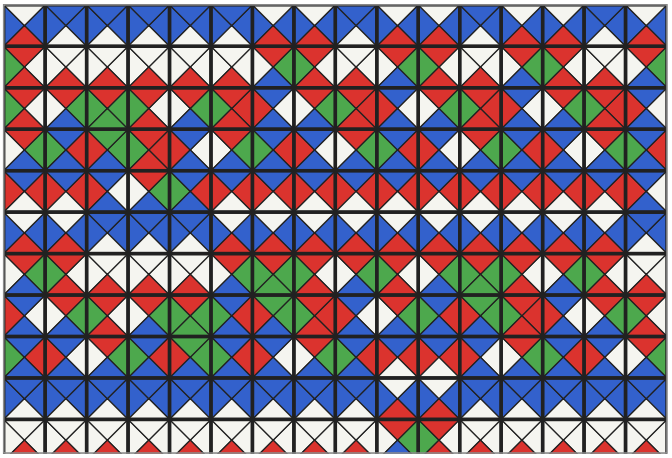
The same pattern, zellij palette

Colouring: by shape · zellij palette

Wang tiles

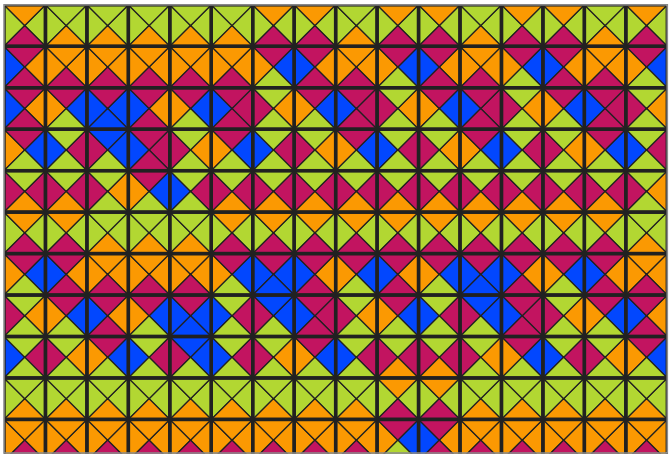
Wikipedia: Wang tile — en.wikipedia.org/wiki/Wang_tile

Squares with coloured edges, placed so that touching edges match and never turned. The 11 tiles of Jeandel and Rao (2015) tile the plane only aperiodically; this patch was found by a backtracking search.



Jeandel–Rao: 11 tiles, 4 colours, aperiodic

Colouring: the four edge colours



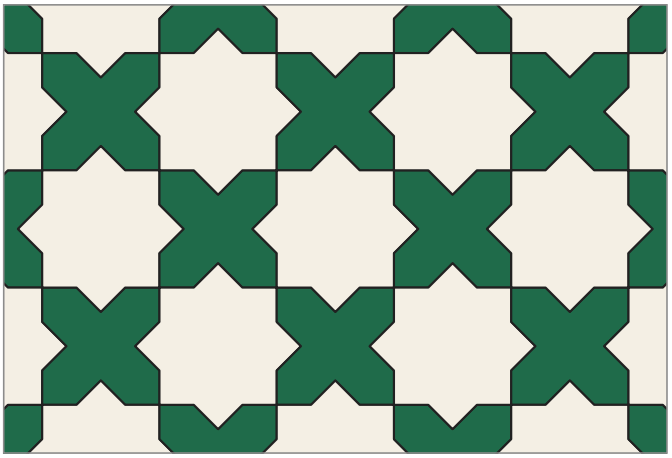
The same tiles, colours from a scheme

Colouring: edge colours from a scheme · square (orange)

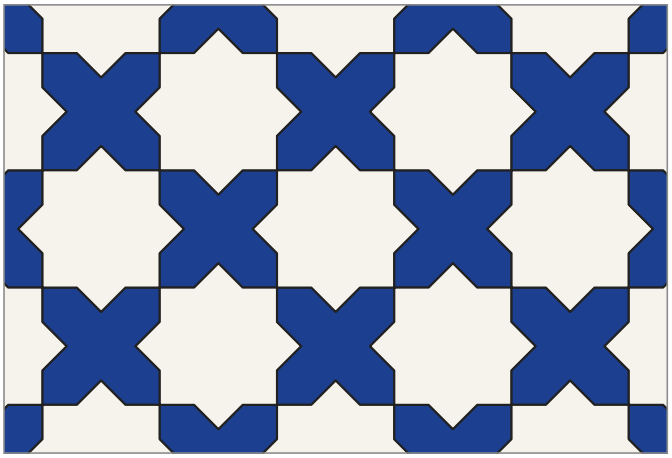
Zellij

Wikipedia: Zellij — en.wikipedia.org/wiki/Zellij

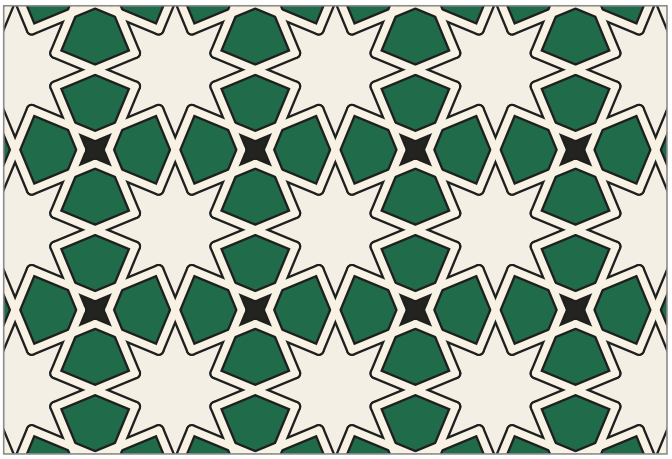
Moroccan and Andalusian mosaic of glazed pieces cut by hand, in white, green, black, ochre and blues. Eight-pointed stars with crosses, and 6-, 8-, 10- and 12-pointed star systems are typical.



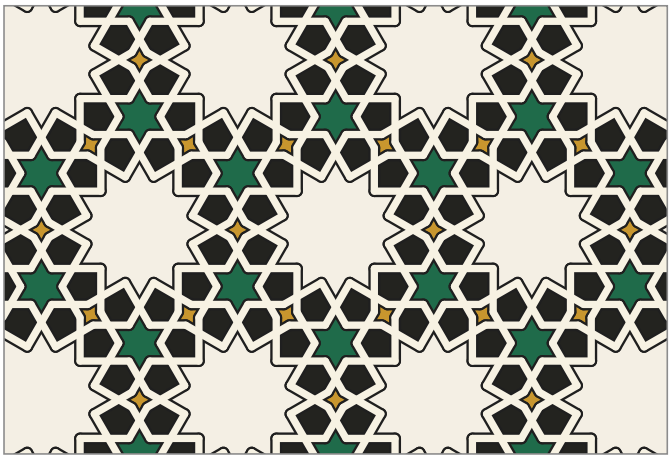
Eight-pointed stars and crosses
Colouring: by kind of tile · zellij palette



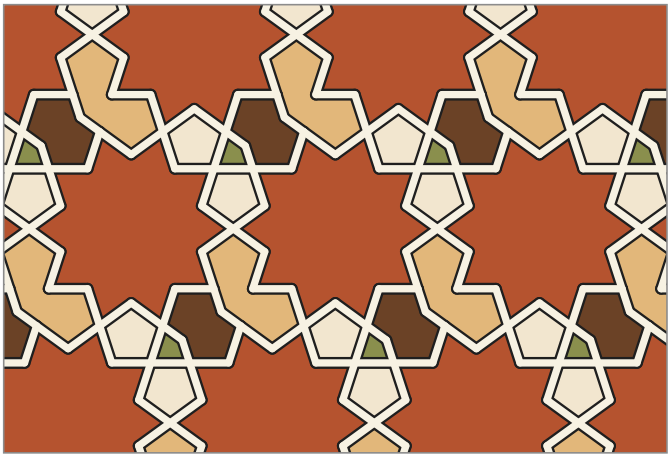
Stars and crosses in azulejo blues
Colouring: by kind of tile · azulejo palette



Eight-pointed stars from octagons in contact
Colouring: by shape · zellij palette



Twelve-, six- and four-pointed stars
Colouring: by shape · zellij palette

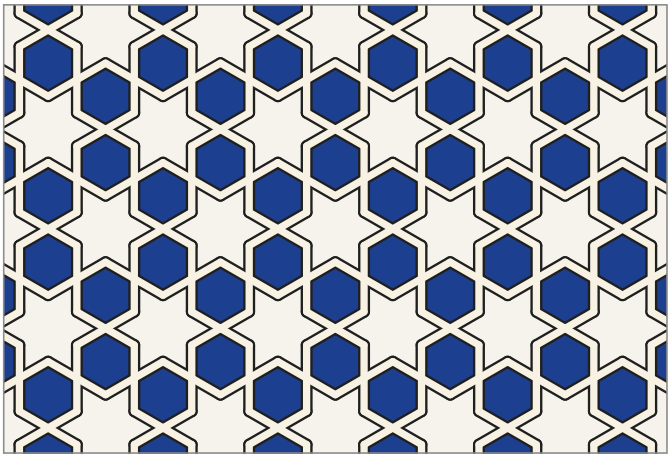


Ten-pointed stars (girih base)
Colouring: by shape · earth tones

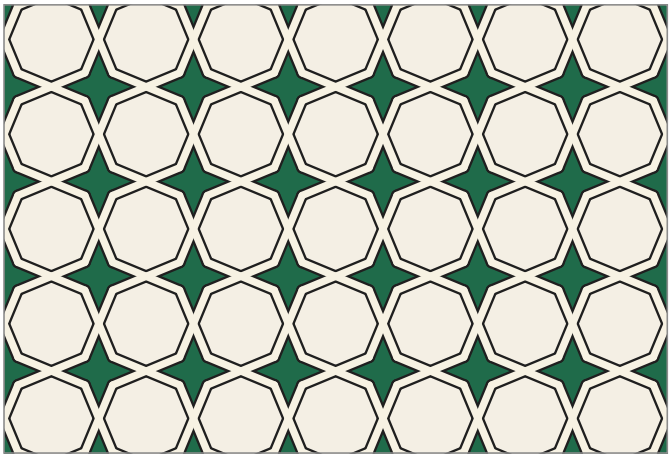
Islamic geometric patterns

Wikipedia: Islamic geometric patterns — en.wikipedia.org/wiki/Islamic_geometric_patterns

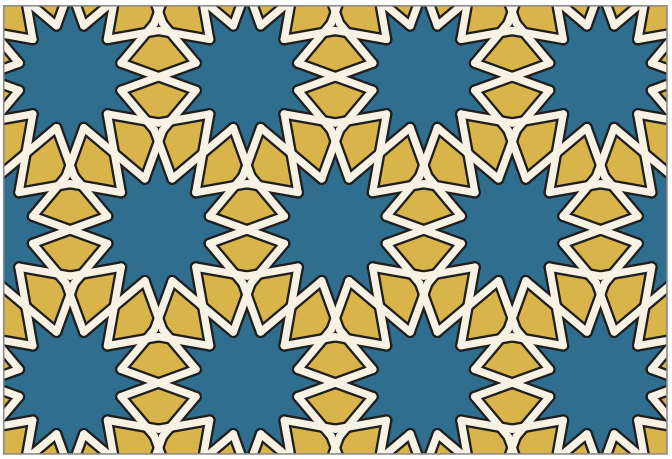
Star patterns made with Hankin's method of polygons in contact: from the middle of each edge of a tiling two lines leave at a fixed angle until they meet. The base tiling and the angle decide the stars; the straps are drawn as ribbons.



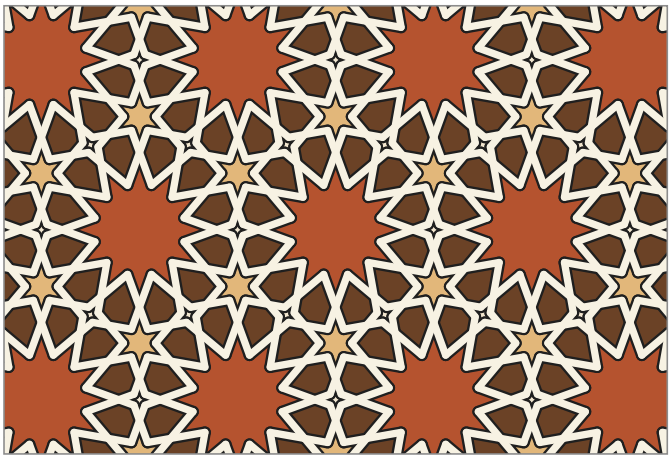
Six-pointed stars from hexagons in contact
Colouring: by shape · azulejo palette



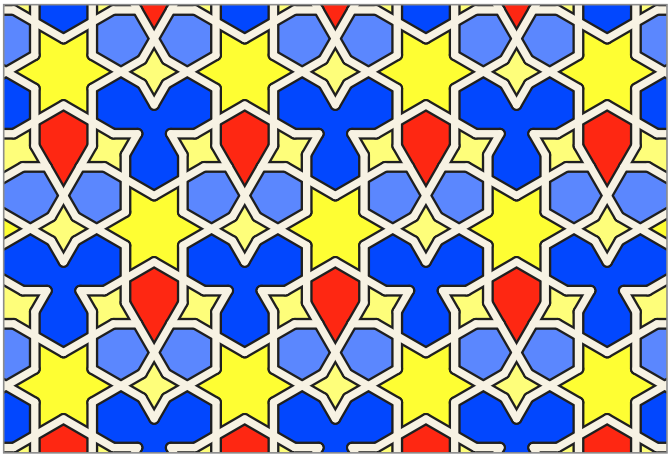
Octagons and four-pointed stars from squares
Colouring: by shape · zellij palette



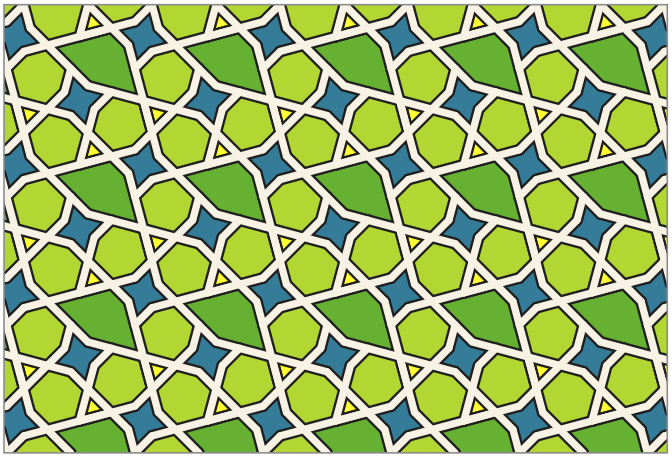
Twelve-pointed stars from 3.12.12
Colouring: by shape · girih palette



Rosettes from 4.6.12
Colouring: by shape · earth tones



Stars from 3.4.6.4
Colouring: by shape · triad (blue)



Stars from the snub square tiling
Colouring: by shape · analogous (green)